

Morphological view on the evolution of the immunity and lymphoid organs of vertebrates, focused on thymus

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Abstract: To understand fundamental defense mechanisms of an individual and to perceive organization and functioning of the human immune system, it is inevitable to sum up its phylogenic development. Lymphoid organs are crucial in regulation of physiological lymphocytes development and subsequently play a significant role in appropriate immunological responses to foreign pathogens. Throughout the evolutionary tree, the primary lymphoid organs have emerged earlier than the secondary lymphoid organs. Considering the sites for cell populations' development responsible for adaptive immunity, B lymphocytes differentiation and maturation have considerably differed during phylogeny as well as ontogeny. On the contrary, T lymphocytes development is defined exclusively in the thymus. From the evolutionary point of view, location of primary lymphoid organs must have been sophisticatedly pre-programmed in terms of their function. Need for thymus evolving from the foregut supports the fact of emerging diverse repertoire of antigen receptors. The thymus represents the very first lymphoid organ evolved in Vertebrata to deal with potentially autoreactive, somatically heterogeneous T lymphocyte receptors. The necessity of maintaining an immunological integrity was the most crucial stimulus for evolution. Thymus as a primary lymphoid organ constitutes an eminent structure that markedly differentiates the higher Vertebrata from the rest of the animal phyla. The present paper is meant to provide a profound evolutionary insight into the lymphoid organs with the emphasis on the thymus morphology through the phylogenesis.

Key words: thymus; lymphoid organs; adaptive immunity; evolution; ontogenesis

Motto

There is hardly any other organ in the vertebrate body about whose functions in the embryo and at later periods there is so little established as it is about the thymus.

J. Beard, University Lecturer in Comparative Embryology, Edinburgh (1900)

Introduction

Homeostasis, as a major attribute of all living organisms, is realized by a row of orchestrated autoregulation processes, including immunity (Sima 1997). Maintenance of the immunological integrity of organisms has been a driving force in evolution. The mechanisms of immunity are evolutionarily selected throughout host – pathogen interactions to be tolerant to self-antigens and to recognize non-self molecular patterns (Rodríguez et al. 2012). During evolution, the thymus emerged concomitantly with adaptive immunity as an “immune brain” highly specialized in the orchestration of central immunological self-tolerance. This was an absolute requirement for the survival of species because of the high risk of autotoxicity inherent to the stochastic generation of extreme diversity characterizing this novel adaptive type of immune defenses against non-self (Geenen 2012a).

The thymus plays a pivotal role in the development of the adaptive immune system, an important factor that separates higher vertebrates from the rest of the animal kingdom (Bowden et al. 2005). The evolutionary biologists often say: “There is no function without a structure, and there is no structure without a history.” The biological history of each organ has two faces, the ontogenesis and phylogenesis. Both are extremely important. The aims of our review are to present a short insight into the evolutionary history of lymphoid organs with a focus on the thymus and speculate on the morphological features of the thymus and other lymphoid organs through the vertebrates' phylogenesis.

Innate and adaptive immunity

Traditionally, all immune mechanisms are subdivided into innate (non-specific immunity) and adaptive (specific-immunity). This division is based on the notion that

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innate mechanisms are encoded in the germ line, while adaptive mechanisms are generated by the recombination and diversification of genes during the development of key cells of the immune system, the lymphocytes. The majority of species is protected only by innate immune mechanisms through the receptors encoded in the genome. On the other hand, the adaptive immune system of vertebrates utilizes receptors created by the rearrangement of gene segments in the primary lymphoid organs (Danilova 2006).

The major functions of the innate immune system include a controlled host – microbial crosstalk at the epithelial barriers, recognition of danger signals, clearance of pathogens by phagocytes, recruitment of mesenchyme-derived professional immune cells and secretion of local or circulating effector molecules such as antimicrobial peptides and simple opsonic forms of complement (Rosenstiel et al. 2009). Phagocytosis, as one of the key mechanisms of innate immunity, is probably the only true principal defense phenomenon common to all the animal kingdom (Šíma & Větvička 1991). About 500 million years ago, jawed vertebrates developed a new system of pathogen recognition. Jawed vertebrates possess a remarkably adaptive immune system that can recognize and initiate a protective response against potentially lethal pathogens. In addition, the adaptive immune system also remembers previous pathogen encounters and can either repel a second invasion or quickly eliminate the recurrent invader by mobilizing a faster and more efficient immune response (Cooper & Alder 2006). The adaptive immune system is defined by lymphocytes (the key cell type of the adaptive immune system), antigen receptors, genes involved in rearrangement, gene conversion and specialized primary and secondary lymphoid tissues or organs (Flajnik & Du Pasquier 2004).

From an evolutionary point of view, the primary lymphoid organs are of special interest because they are involved in the generation of the primary repertoire of antigen-specific receptors expressed on lymphocytes (Boehm & Bleul 2007). In all jawed vertebrates, the thymus is the major site for T lymphocyte maturation. On the other hand, the sites for B lymphocyte development are less clear and B lymphocyte development can occur in a variety of anatomical sites. In elasmobranchs, the spleen and Leydig's organ (a hemopoietic organ located in the submucosal layer of esophagus) both contribute to B lymphocyte development. In teleosts, the kidney is undeniably the major source of B lymphocyte development. Amphibians appear to use several different sites (spleen, bone marrow and/or kidney) depending upon the species (Hansen & Zapata 1998). In birds, the *bursa cloacalis* (of Fabricius) is critical for the normal development of B lymphocytes (Ratcliffe 2006; Dorko et al. 2013). Mammalian B lymphocytes during embryogenesis begin their development in the yolk sac (Butler et al. 2006) or fetal liver and during postnatal phases in the bone marrow. In humans and rodents, B lymphocytes development and primary B lymphocyte repertoire diversification occurs in the bone

marrow. However, in some species the populations of B lymphocytes complete their maturation only after migration into the spleen (e.g., mouse) (Boehm & Bleul 2007; Brendolan et al. 2007) or also in the lymph nodes (e.g., in bovine) (Ekman et al. 2010). In recent years it has become clear that the gut associated-lymphoid tissue (e.g., aggregated lymphoid nodules of the aboral part of ileum, called Peyer's patches), probably a mammalian equivalent of the avian bursa of Fabricius, is also critically important for the B lymphocyte development in many species (e.g., sheep, cattle, rabbits and pigs) (Alitheen et al. 2010; Ratcliffe 2002; Yasuda et al. 2006). These preliminary results suggest that the gut-associated lymphoid tissue could be considered as a primary lymphoid organ in some mammals.

Quick morphological overview of the thymus

The thymus is the most ancient primary lymphoid organ known and a component of the adaptive immune systems of all vertebrates (Boehm 2011) (Table 1). It also appears to be an obligatory intersection for the integrated evolution of the major systems of cell-to-cell signaling and forms a special crossroad between the nervous, endocrine and immune systems (Mravec et al. 2009; Geenen 2012a). Since its first description by Claudius Galenos of Pergamon (130–200 AD), the thymus has remained an “organ of mystery” throughout the 2,000 years of history of biology and medicine (Kachlik et al. 2008). The general description of the thymus is complicated. The size and microscopic structure of the thymus is variable and depends on the age of the individual. One of the most characteristic changes of the ageing immune system is the regression, or involution of the thymus. Moreover, age-associated thymic involution seems to occur in all vertebrates that possess a thymus, indicating that this is an evolutionary, ancient and conserved event. Age-related thymic involution (sometimes shortly described as a “natural shrinking process”) is associated with a reduction in tissue mass and thymic cellularity as well as loss of tissue structure and abnormal architecture. Although it is accepted that many factors contribute to age-associated thymic involution, little is known about the mechanisms involved in the process (Aw & Palmer 2011; Rezzani et al. 2014). The thymus also changes its structure under exogenous negative stimuli (e.g., hormones, pesticides, certain diseases, poor nutrition, stress or radiotherapy) (Varga et al. 2011b; Dorko et al. 2011a, b). Even though thymopoiesis is maintained until late in life, thymic involution nevertheless remains the hallmark of immunosenescence that is characterized by a higher susceptibility to infections as well as a decrease in the vaccine and anti-tumor immune responses (Geenen 2012b).

The thymus plays an important role in providing a suitable microenvironment for the proliferation, differentiation, T lymphocyte receptor gene rearrangement and repertoire selection of **T lymphocytes** (Aifantis et al. 2006; Rezzani et al. 2008; Manley et al. 2011).

Table 1. Summary of the lymphoid organs evolution in Vertebrata.

Taxonomic group	Thymus	Bone marrow	Spleen	Lymph nodes and lymph vessels	Other lymphoid organs
Jawless vertebrates	Thymus-like tissue at the tip of gill filaments ("thymoid")	No	No	No	Larval period: typhlosole, liver, opisthonephros Adults: kidney, supraneural body
Cartilaginous fish	Yes, in some species also with a duct	No	Yes	In general no, only some species have lymph vessels	Gut-associated lymphoid tissue, Leydig's organ in oesophagus, epigonadal organ and cranial lymphoid tissue in chimaera
Bony fish	Yes	No	Yes	Only lymph vessels	Melano-macrophage centers in kidney, liver and spleen, GALT interbranchial lymphoid tissue
Amphibian	Yes, typically season-dependent morphological changes	Only in Anura and terrestrial Urodela during adulthood	Yes	Lymph vessels with lymphatic hearts	Jugular bodies, GALT, in <i>Gymnophiona</i> lymphopoiesis occurs in kidney and liver
Reptiles	Yes	Yes	Yes	Lymph vessels with lymphatic hearts, in some species primitive lymph nodes in axillary and inguinal regions	GALT – intestinal and cloacal lymphoid aggregates
Birds	Yes	Yes	Yes	Lymph nodes (only in some species), lymph vessels and in some species also lymphatic hearts (mostly <i>in ovo</i>)	Bursa of Fabricius, Harderian gland, GALT include caecal tonsils, Peyer's patches and Meckel's diverticulum
Mammalian	Yes	Yes	Yes	Yes (in some species also haemal lymph nodes)	Tonsils, Peyer's patches in ileum

The differentiation of T lymphocytes is a carefully orchestrated process supported by the thymic microenvironment, composed of thymic epithelial cells, fibroblasts, myoid cells and bone marrow-derived accessory cells such as B lymphocytes, macrophages and interdigitating dendritic cells. The presence of non-lymphoid cells through the thymic parenchyma serves to provide a coordinated migration and differentiation of T lymphocytes (Rezzani et al. 2008; Jablonska-Mestanova et al. 2013). During the immunocompetence acquiring process high number of developing lymphocytes die due to inappropriate reactions to self as well as non-self antigens. These lymphoid cells demonstrating altered reactivity are eliminated in a way of programmed cell death – **apoptosis** (Bai et al. 2013; Konkel et al. 2014). Dead lymphocytes are most frequently removed by macrophages (Saranq et al. 2013). Apoptosis is an essential process in organism of maintaining the general homeostasis occurring not only under the physiological conditions including normal cell turnover, proper development and functioning of the immune system (Elmore 2007; Paganelli et al. 2008) but, during continuous pathological processes, e.g. neurodegenerative diseases, ischemic damage or carcinogenesis (Adamkov et al. 2011, 2014; Halašová et al. 2010; Kovalska et al. 2012).

The thymus of mammals is a soft, encapsulated and lobulated organ. The capsule and septa consist of connective tissue that supports blood, lymphatic vessels and nerve fibers. The septa divide the parenchyma into incomplete lobules, each of which contains a cortical and a medullar part. Stained histological sections show a **darker cortex** coating a **paler medulla** (Fig. 1). The cortex of the thymus lobes is darker due to the presence of numerous closely packed maturing T lymphocytes (Raica et al. 2007; Varga et al. 2009). Since 1849, when a London physician and chemist Arthur Hill Hassall described the acidophilic squamous spherical structures in the thymic medulla, these corpuscles have been thought to be specific for this organ. **Hassall's bodies** are in fact multicellular swirled epithelial structures, variable in number and size, often displaying degenerative changes in the central area, such as necrosis, cellular detritus and sometimes even extensive calcification or cystic alterations (Raica et al. 2006; Varga et al. 2010). They not only have a distinctive histological appearance inside one species, but also different occurrence and structure throughout different animal species. For example, they are typically found in human (Fig. 2), rats, guinea-pig or chicken thymuses, but are not so typically present in mice or elasmobranchs (Bodey et al. 2000; Hale & Markert 2004; Raica et al. 2006; Kannan

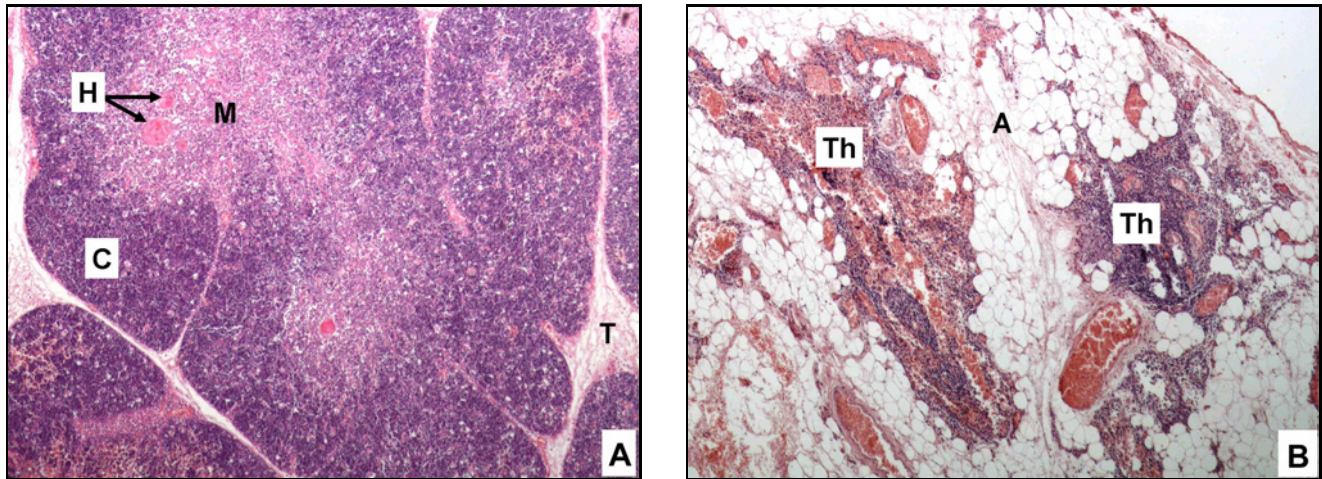


Fig. 1. Microscopic structure of human thymuses in two different age groups. A: Thymus of 2-month-old boy (C – cortex, M – medulla, H – Hassall's bodies, T – trabecula from connective tissue). B: Thymus of 24-year-old woman (A – adipose tissue, Th – remnants of thymic tissue) (hematoxylin & eosin, 40 $\times$).

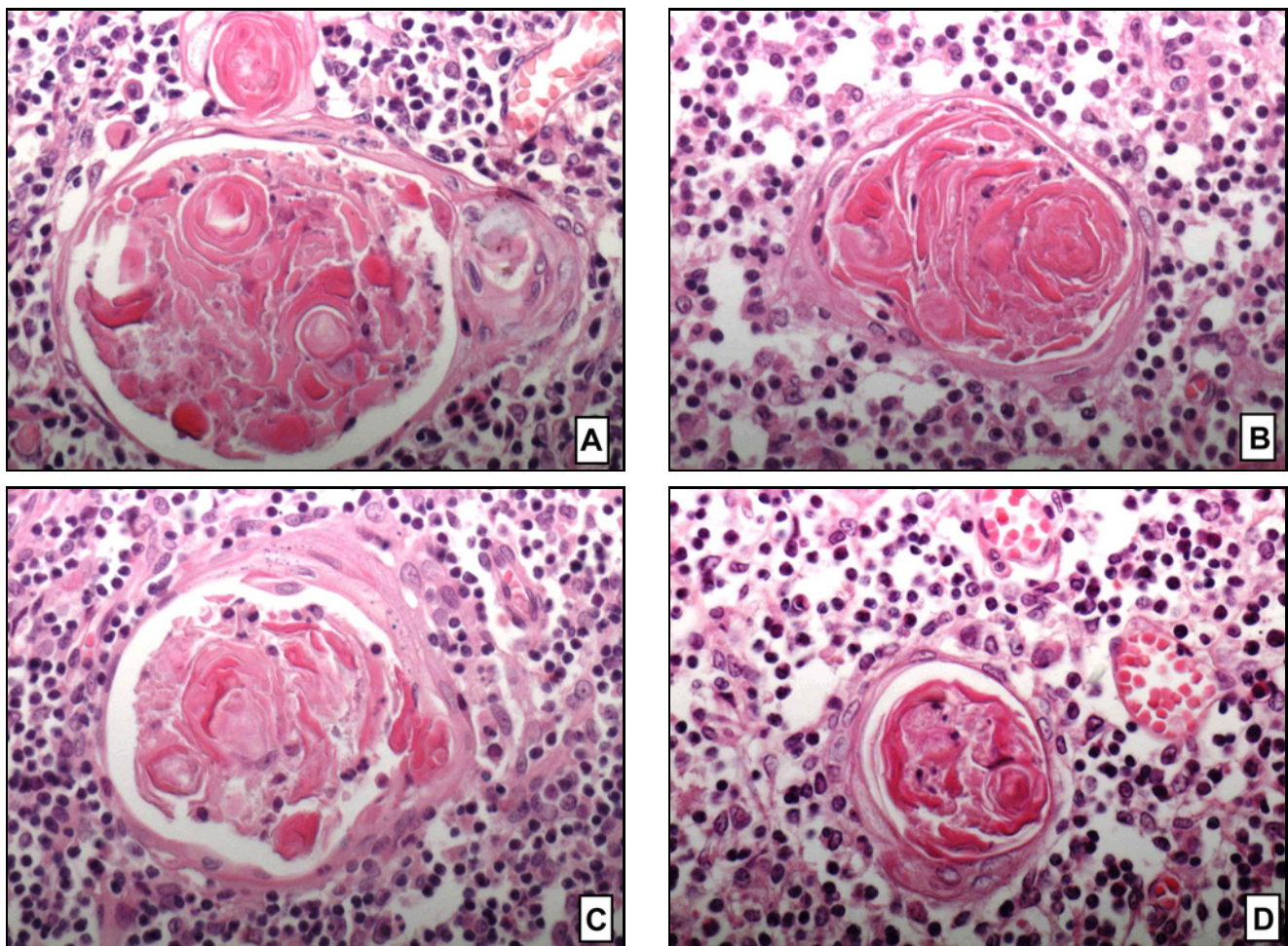


Fig. 2. Examples of morphologically distinct Hassall's bodies from thymuses of children in age from 1-week up to 4-month (hematoxylin & eosin, 400 $\times$).

et al. 2015; Valančiūtė et al. 2015). Hassall's bodies have been shown to be involved in the regulation of lymphocytes and epithelial cells maturation (Nishio et al. 2001) as well as to play a role in the communication between antigen-presenting cells and developing T lymphocytes (Zaitseva et al. 2002).

The most important non-lymphoid cell population is represented by morphologically and functionally different **thymic epithelial cells** forming the cellular network (Fig. 3). The epithelial cells in the cortical part are often referred to as “**thymic nurse cells**” due to the fact that they provide a unique microenvironment

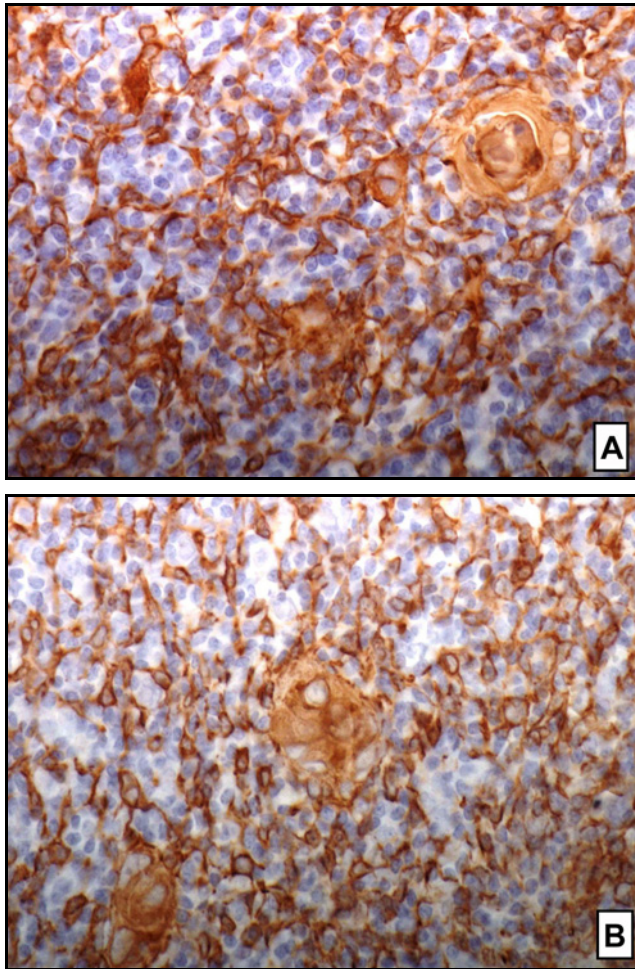


Fig. 3. Cellular network formed by a processes of thymic epithelial cells in thymus of 1-year-old girl. A: Antibodies anti-cytokeratins AE1/AE3. B: Antibodies anti-high molecular weight cytokeratins (visualized with diaminobenzidine, cell nuclei stained with Mayer's hematoxylin; 400 $\times$).

in the thymus for T lymphocyte repertoire development and selection (Nakagawa et al. 2012; Reyes García & García Tamayo 2013).

Other non-lymphoid cell populations within the thymus are derived from the mesenchyme (from hemopoietic stem cells) and include T lineage cells as the dominant cell type, together with a low level of B lymphocytes, macrophages and interdigitating dendritic cells. The **macrophages** participate mainly in the removal of apoptotic lymphocytes (Fig. 4) (Wakimoto et al. 2008). However, a less accepted hypothesis states, macrophages implication in the regulation of the survival, proliferation, or negative selection of developing lymphocytes (Varas et al. 2003). **Dendritic cells** form a dynamic cellular network (Fig. 5) and also possess an important immune-regulatory function, besides their notorious antigen-presenting capacity (Chung et al. 2013). The thymus probably also contains peripheral dendritic cells, which work as cellular transporters of peripheral tissue-specific antigens or foreign innocuous antigens important for the development of an effective central tolerance (Hadeiba & Butcher 2013).

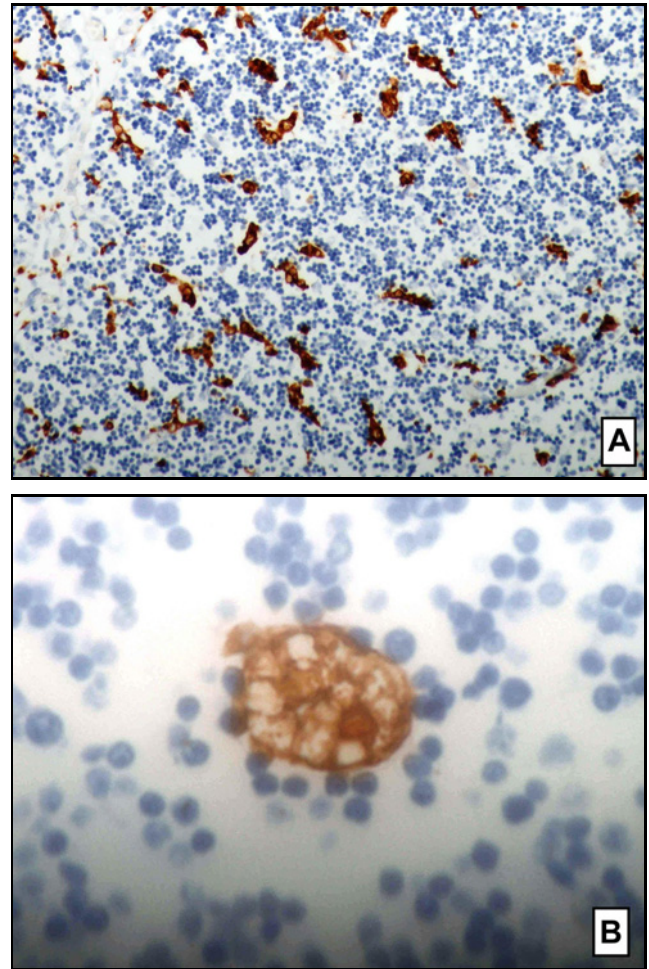


Fig. 4. Thymic macrophages visualised with anti-CD68 antibodies and final reaction stained with diaminobenzidine. Cell nuclei stained with Mayer's hematoxylin. A: Macrophages in thymic cortex of 2-month-old boy (200 $\times$). B: Inside cytoplasm of macrophage numerous lymphocyte-like cells (1,000 $\times$).

Inside the thymic medulla we can also find cells similar to the striated muscle cells (Fig. 6). These **thymic myoid cells** were first noted in 1888 by Mayer, especially in the frog thymus (Mayer 1888). Until today, the embryonic origin and exact role of the myoid cells in thymic physiology is not clear, but they are probably derived from the neuroectoderm of the neural crest (Nakamura & Ayer-Le Lière 1986). The most acceptable hypothesis on the myoid cells function is that they protect the developing lymphocytes from apoptosis as well as play an important role in the differentiation of the T lymphocytes (Panse & Berrih-Aknin 2005).

The lymphoid “organs” and thymus-analogous organ in jawless vertebrate

The first jawless vertebrates appeared in the Cambrian period some 450–500 million years ago. However, lamprey and hagfish are the only groups of agnathans that have survived to the present. The jawless vertebrates (superclass Agnatha) comprise of only about one hundred species. Nevertheless, it is their unique evo-

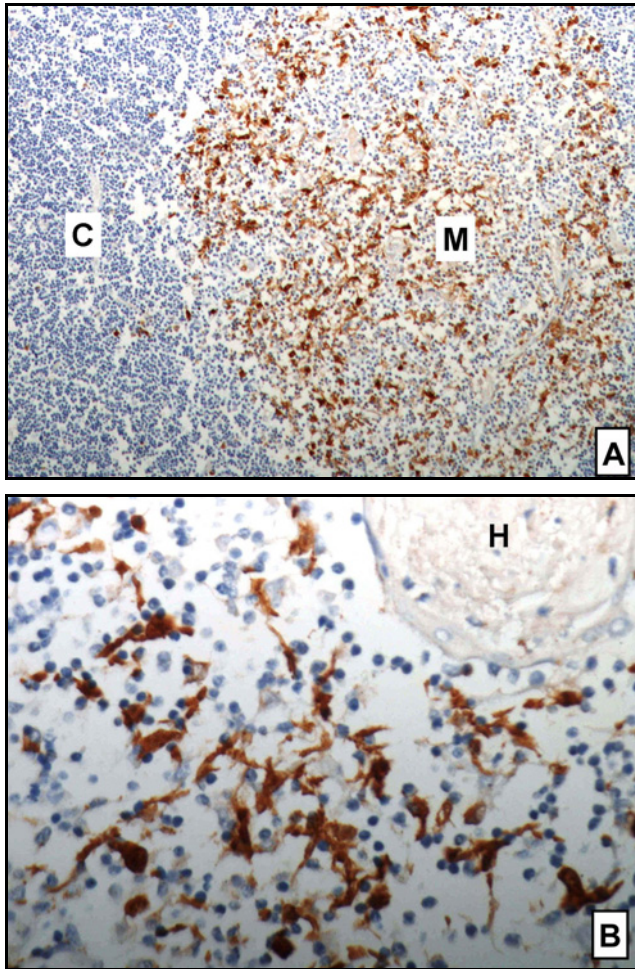


Fig. 5. Interdigitating dendritic cells of thymus visualised with anti-S100 antibodies and final reaction stained with diaminobenzidine. Cell nuclei stained with Mayer's hematoxylin. A: Cellular network formed by dendritic cells in thymic medulla of 3-month-old boy (C – cortex, M – medulla; 100 $\times$). B: Dendritic cells with numerous thin cytoplasmic processes in thymic medulla of 2-month-old boy (H – Hassall's body; 400 $\times$).

lutionary position as the most primitive vertebrates why they attract the interest of evolutionary and developmental biologists (Osório & Rétaux 2008; Boehm 2011). The jawless vertebrates express some progressive immunological features and are the first animals which have been able to produce immune cells similar to both B and T lymphocytes (Sima & Vetvicka 1993).

The morphology of immune system is most studied in the **sea lamprey** (*Petromyzon marinus* L., 1758). The sea lamprey has a long life cycle, during which it undergoes a complex metamorphosis. During the larval period, the worm-like larvae (*ammocoetes*) remain buried in the river silt, where they filter feed on plankton. Later on during metamorphosis, lampreys undergo an extensive modification affecting their whole anatomy and physiology, and subsequently, the adults parasitize fish. Unlike the larvae, the adult lampreys have teeth arranged in a concentric circle, which they use to attach to fish and feed on their body fluid (Osório & Rétaux 2008; Herrin & Cooper 2010).

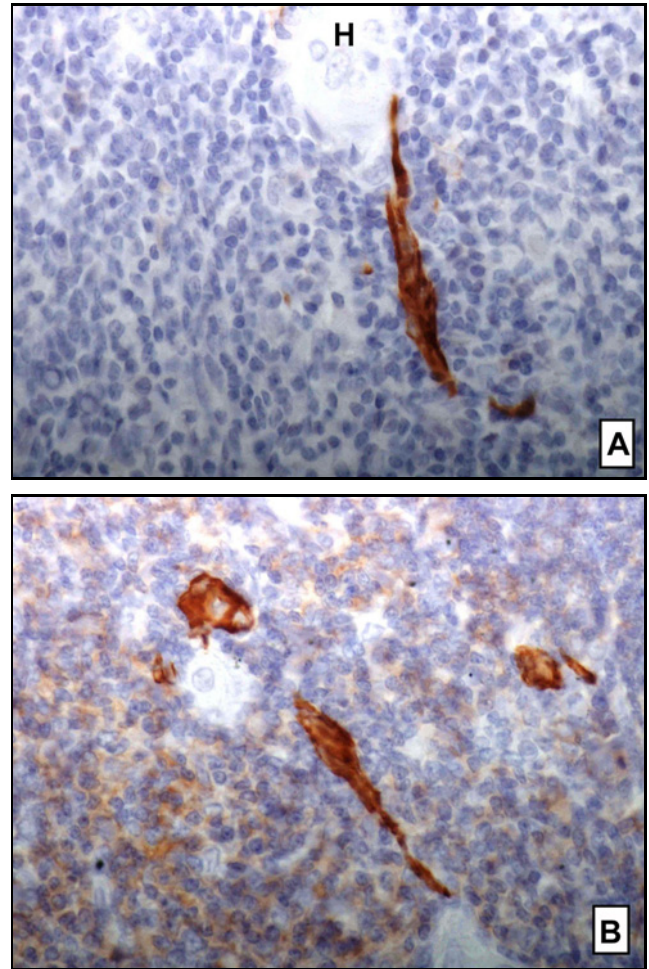


Fig. 6. Elongated thymic myoid cells from thymic medulla of 1-year-old girl. A: Antibodies anti-desmin. B: Antibodies anti-alfa-actin (visualized with diaminobenzidine, cell nuclei stained with Mayer's hematoxylin; H – Hassall's body; 400 $\times$).

The lampreys have **no true lymphoid organs** and, accordingly, their immune responses seem more related to the non-anticipatory defense mechanisms of invertebrates than to the immune responses of vertebrates (Zapata et al. 1995). On the other hand, the **lymphocyte-like cells** of the lampreys are morphologically similar to T and B lymphocytes of jawed vertebrates. These lymphocyte-like cells clonally express somatically diversified antigen receptors termed variable lymphocyte receptors (VLR), which are composed of tandem arrays of leucine-rich repeats. VLRA is expressed on the surface of T like lymphocytes, while B like lymphocytes express and secrete VLRA as a multivalent protein (Boehm et al. 2012b). The immunization of lampreys with bacteria, viruses or mammalian cells induces the lymphocyte-like cells to undergo transformation, proliferation and differentiation into antibody-secreting cells. These cells are ultrastructurally similar to the higher vertebrates' plasma cells (they have an extensive cytoplasm with abundant rough endoplasmic reticulum) in jawed vertebrates (Herrin & Cooper 2010). Agnathans are the first animals able to react to an antigenic challenge by the production of antibodies.

Most of the lampreys' lymphoid tissue is located very close to the rostral part of the digestive tube exposed to exogenous antigens (Varga et al. 2008). Lympho-hematopoietic cells during larval period occur mostly (Ardavin et al. 1984, Fujii 1982; Ardavin & Zapata 1987):

In the pharyngeal region (gill basket). According to some authors, numerous blood sinuses and accumulations of lymphocytes and macrophages in the larval pharynx can be similar to the thymus in jawed vertebrates (so-called "thymoid", discussed later);

Esophagus and anterior intestine. The intestine contains also a special invagination of the dorsal wall (*typhlosole*), which enlarges the gut surface. *Typhlosole* is the first place of hemopoiesis during development and can be an analogous organ to the later spleen. During metamorphosis, the *typhlosole* is embedded in the dorsal wall of the intestine growing toward the cloaca;

Liver;

Nephric fold (larval *opisthonephros*).

After metamorphosis, the *typhlosole* regresses and hemopoiesis occurs in the intertubular spaces of an adult kidney and supraneural body (a so-called protovertebral arch or dorsal fat body) (Ardavin et al. 1984). From a morphological point of view, the supraneural body appears similar to the bone marrow in higher vertebrates, since all types of blood cells in all stages of maturity, as well as their precursors, are present in this tissue (Amemiya et al. 2007).

The search for a **thymus-analogous organ** in the lamprey was based on the presumption that all vertebrates possess and require this structural innovation for immune function (Ardavin et al. 1984; Saha et al. 2010). In the past, most evolutionary biologists negated the presence of thymus in lampreys. For example, in 1894 Beard wrote: "Above the gills of the adult lamprey, of the *ammocoete*, and very young larval forms no thymus is to be found. At present I incline to the belief that the Marsipobranchii (alternative name for Cyclostomata) possess no well-defined thymus." Nevertheless, light microscopic studies identified a thymus-similar tissue in the lamprey's larvae in association with the respiratory pharynx, on the tips of gill filaments. Šterba (1953) described an epithelial reticulum resembling the thymic epithelial cells as well as a selective accumulation of lymphocytes in this area. But is this pharyngeal lymphoid tissue a thymus-analogous organ or not? The first hypotheses predicted that this dispersed lymphoid tissue localized under the pharyngeal epithelium appears to be involved mostly in the uptake of foreign materials from the pharyngeal cavity (Page & Rowley 1982). According to Amemiya et al. (2007) this lymphoid tissue clearly was neither thymus nor a primordial precursor of such an organ. On the other hand, modern methods of molecular immunology discovered a site of VLRA expressing T like lymphocytes development in lampreys. The site of their development is in a thymus-like tissue at the tip of gill filaments (so-called "**thymoids**"), under the pharyngeal epithelium. These studies confirm that dispersed thymoids local-

ized in the pharyngeal region of lampreys are, indeed, **thymus-analogous organs** (Guo et al. 2009; Bajoghli et al. 2011; Boehm et al. 2012a, b). The finding of T like and B like lymphocytes in lampreys offers new insight into the evolution of adaptive immunity. The thymoids as a "primitive thymus" based on their structure, function and development are probably not only thymus-analogous organs. There is a **true homology** between lamprey's thymoids and the thymus of higher vertebrates (Ge & Zhao 2013).

Thymus and other lymphoid organs in cartilaginous fish

Cartilaginous fish (superclass Chondrichthyes) occupy a fundamental position in vertebrate phylogeny and it is likely that this group has retained some of the ancestral immune mechanisms. Elasmobranchii, a subclass of Chondrichthyes (cartilaginous fish), are jawed fish that include modern sharks, rays, guitarfish and skates. Cartilaginous fish, regardless of their rather archaic and simple vertebrate body pattern, are the first vertebrates characterized by well-developed cellular and humoral immune mechanisms and inside lymphoid organs well organized zones containing B and T lymphocytes are found (Šíma & Větvička 1991). Elasmobranchs lack bone marrow and lymph nodes, but possess a thymus (at least in early life), spleen, gut-associated lymphoid tissue and important lympho-hemopoietic organs associated with gonads (as epigonal organ), esophagus (as Leydig's organ), or both. The epigonal and Leydig's organs both have similar histological structure and resemble the bone marrow of the higher vertebrates (Fänge & Mattisson 1981; Honma et al. 1984).

The elasmobranch **thymus** is situated dorsomedially to both gill regions. Its size and shape varies with sexual maturation. Separate cortical and medullary regions resemble the cortex and medulla of the mammalian thymus, except for the lack Hassall's bodies in the medullary region. The parenchyma consists mostly of lymphocytes ranging in size from small, darkly staining cells to large, lightly staining blast-like cells (Luer et al. 1995). In some species, the thymus may have an excretory character and is joined with the branchial clefts via a duct. The thymus in shark *Heptanchus* appears to be very interesting from a morphological point of view. During its development, all thymic primordia are claimed to be joined with the clefts through an epithelial tubule (Slípka 1986). Also developing thymus of electric ray *Torpedo torpedo* (L., 1758) form short duct and resembles exocrine gland (Pospíšilová et al. 2003). According to Slípka (1986), this structure confirms that the thymus once had an excretory character and developed as a modification of the *branchionephros* of the marine invertebrate *Amphioxus* sp. The thymus is a typical organ derived from the branchial region. In elasmobranchs, the branchial region reaches its highest structural and functional stage of development. In sharks, this region contains the most parts of circulatory system including the heart as well as the first real

thymus developed here. The number of its primordia corresponds to the number of branchial clefts (Slípka 1986), but the development of a real thymus depends on the species.

The spleen of elasmobranchs is an important lymphomyeloid organ and consists of well-defined red and white pulp. The sharks' white pulp contains a central T lymphocyte zone with dendritic cells, while the red pulp contains cells of erythropoietic line mixed with other white blood cells (Fänge & Mattisson 1981; Rumfelt et al. 2002).

The gland like structure in the wall of esophagus of elasmobranchs was first discovered by a German comparative anatomist Franz von Leydig (1821–1908), who interpreted it as a lymph node. In some elasmobranchs, the Leydig's organ is remarkably large, e.g., 1.6 kg in one 1.8 m long cow shark (*Hexanchus corinus* Jordan and Gilbert, 1880), but in some other species this organ completely lacks. **The Leydig's organ** consists of higher vertebrates' bone marrow-like tissue situated between the mucosa and tunica muscularis of the esophagus (Mattisson & Fänge 1982). The main function of the Leydig's organ is the formation and storage of white blood cells (especially granulocytes) and after experimental spleen extirpation also the formation of red blood cells (Oguri 1983).

The main functions of **the epigonal organ** are the production and storage of white blood cells and being the major site of neonatal B lymphopoiesis (Rumfelt et al. 2002). The epigonal organ of the Atlantic nurse shark (*Gynglymostoma cirratum* Bonnaterre, 1778) is a Y-shaped structure located intraperitoneally in the abdomen. In males, the two testes are embedded within the left and right anterior part of the epigonal organ, while in females; the right anterior part of the epigonal organ is enlarged and contains the right ovary. Microscopically, the epigonal organ contains white blood cells in various stages of development, in the network of connective tissue (Fänge & Mattisson 1981). The weight of the epigonal organ is similar to the weight of the spleen and formed 0.2–0.6% of total body weight (in comparison, the Leydig's organ formed only 0.05–0.15% of total body weight) (Fänge & Pulsford 1983). Lutton & Callard (2008) demonstrated also functional interactions between gonads and the epigonal organ in the skate, because the cellular turnover in epigonal tissue is influenced by gonadal activity.

Little is known about the elasmobranchs' **gut-associated lymphoid tissue**, which contains mostly lymphocyte and macrophage-like cells in the gut's lamina propria. The size of lymphoid accumulations in the lamina propria increases when the gut is exposed to exogenous (food) antigens (Hart et al. 1986).

Holocephali is another subclass of Chondrichthyes, evolutionary older than the previous, of which the order Chimaeriformes is the only surviving group. It includes the rat fish (e.g., *Chimaera*), rabbit-fish (e.g., *Hydrolagus*) and elephant-fish (e.g., *Callorhynchus*). Holocephalans have a spleen and a thymus, similar to elasmobranchs. In species of *Chimaera* a voluminous lym-

phomyeloid tissue is found in the cranium, mainly in the orbit, the preorbital canal, and the suprapalatal area (Fänge & Sundell 1969). The presence of macrophages, lymphocytes and developing and mature plasma cells suggests an important role of the **cranial lymphohemopoietic tissue** in eliciting the immune responses. A phylogenetic relationship between this tissue and the higher vertebrate bone marrow is proposed on the basis of histological similarities between the cellular microenvironments governing haemopoietic differentiation in these organs (Mattisson et al. 1990). *Chimaera monstrosa* L., 1758 have a well-developed thymus, which probably does **not undergo any age-related involution** (Fänge 1977).

Thymus and other lymphoid organs in bony fish

Bony fish (superclass Osteichthyes) are a group of fish that have bony, as opposed to cartilaginous, skeletons. Osteichthyes are divided into the ray-finned fish (class Actinopterygii), including most of bony fish and lobe-finned fish (class Sarcopterygii), including lungfish and coelacanths. The class of ray-finned fish is the largest vertebrate class – they make up about half of all known vertebrate species. They comprise of almost 28,000 described species with considerable morphological and physiological variations. The most living fish are members of the Teleostei group and also the lymphoid system is described in teleosts in most detail (Schröder et al. 1998; Picchietti et al. 2009; Patel et al. 2009; Koppang et al. 2010; Øvergård et al. 2011).

The major lymphoid organs of teleosts are the thymus, bone marrow-analogue located in the kidney (pro- and mesonephros), spleen (Chantanachookhin et al. 1991), and a newly discovered diffuse mucosal gill-located interbranchial lymphoid tissue (Dalum et al. 2015).

The thymus in teleosts was discovered by Staninus in 1850. In most teleosts, the thymus is located near the gill cavity and is closely associated with the pharyngeal epithelium (Bowden et al. 2005). In some species, such as angler fish (*Lophius piscatorius* L., 1758), the thymus is located behind the gill cavity and is separated from the pharyngeal epithelium by a layer of loose connective tissue. Also in angler fish, a duct-like tube with epithelial lining containing a number of large mucous cells extends from the pharynx to the thymus (Burne 1926). The gross anatomy of the thymus in teleosts varies among species. The thymus can appear as more than one pair of bilobed organ with a conical shape. For example, in the cling fish (*Sicyases sanguineus* Müller & Troschel, 1843), each gill chamber has a pair of thymus with one taking up a superficial position and the other located close to the gill epithelium (Gorgollon 1983). In other species, e.g. in zebra fish or carp, the thymus forms characteristic protrusions dorsally among the skull during growth (Danilova et al. 2004). The age-related involution, which characterizes the thymus of the higher vertebrates, does not necessarily occur in bony fish. Nevertheless, thymus func-

tion may be modulated by season, maturity, and stress (Chilmonczyk 1992).

The microscopic structure of the thymus consists of differentiating lymphocytes within a network of epithelial cells. The parenchyma is generally organized in **cortical- and medullary-like zones**, however the presence of a “cortex” and “medulla” is not a constant feature of all fish species (Romano et al. 1999a, b). According to Koppang et al. (2010), based on immunohistochemical analysis of sexually immature salmonids, the anatomical organization of the thymic cortex and medulla seems to be composed of three layers consisting of a sub-epithelial medulla-like zone, an intermediate cortex-like zone and finally another cortex-like basal zone. The thymus of some species contains concentric structures resembling the Hassall’s bodies (Prymak 1902; Bowden et al. 2005). Thymuses of some species may contain also cystic cells (Gorgollon 1983). Immunohistochemically, in thymuses of juvenile turbot (*Psetta maxima*, 1758) also neuroendocrine cells were described in the inner zone of the thymus in close association with lymphoblasts and lymphocytes (Vigliano et al. 2011).

The **interbranchial lymphoid tissue** is quantitatively a very important site of T lymphocytes aggregation, strategically located to facilitate the antigen encounter (Koppang et al. 2010; Austbø et al. 2014; Dalum et al. 2015). The microscopic structure of the **spleen** in the *Cyprinus* species differs from the mammalian spleen and contains a ring of red pulp under the capsule and white pulp in the center (Tischendorf 1985). In freshwater teleosts, the thymus is the first organ to become lymphoid, although the kidney may contain hematopoietic progenitor cells, but no lymphocytes, even earlier. However, in marine teleosts, the order in which the major lymphoid organs develop is kidney, spleen and, finally, the thymus (Zapata et al. 2006). In most studied teleosts, B lymphocytes have been suggested to originate and differentiate in the kidney, but the origin of lymphoid progenitors that colonize the developing thymus and developing into T lymphocytes is unknown (Hansen & Zapata 1998). The **kidney** (pronephros and mesonephros), as well as the liver and spleen of some species may contain **melanomacrophage centers**, also known as macrophage aggregates, usually containing the pigments hemosiderin, lipofuscin, ceroid, and melanin (Wolke 1992). Melanomacrophage centers increase in size or frequency in conditions of environmental stress or inflammation and have been suggested as reliable biomarkers for water quality in terms of both deoxygenation and chemical pollution (Agius & Roberts 2003).

From the evolutionary point of view, **lungfish** have an interesting phylogenetic position. Increasingly, molecular evidence indicates that lungfish are the closest extant group of fish to tetrapods (Joss 1998). In juvenile Australian lungfish (*Neoceratodus forsteri* Kreft, 1870) the thymus is a bilateral organ that lies anterior and dorsal to the most anterior gill arch. The thymic microscopic structure shows clear demarcation into cortex and medulla. In contrast to

the cortex, the medulla has relatively few lymphoid cells. Just like in thymuses of other bony fish, epithelial cells form a network both in cortex and medulla. The thymic tissue of lungfish further contains macrophages, myoid cells, secretory-like cells (similar to mast cells) and large cells including foamy and giant multinucleated cells (Mohammad et al. 2007).

Thymus and other lymphoid organs in amphibians

Class Amphibia gives name to the cold-blooded animals, which have a special position in Vertebrata subphylum, since they are first representatives of Tetrapoda, the four-limbed vertebrates (Duellman & Trueb 1986). Description of lymphoid system and studying ontogeny of immunity in amphibians is a demanding task linked to the aspects of complicated embryonic and larval development and subsequent morphological changes the members of this order undergo. Metamorphosis in amphibians presents a unique problem for the developing immune system. Because of the fact, tadpoles are free-living organisms; their immune system must inevitably possess the ability of protection against potential pathogens. During metamorphosis, tadpoles’ immune system has to bear with the complex process of acquiring variety of new adult-specific molecules to which tadpoles must evolve an immunological tolerance (Du Pasquier 1982; Rollins-Smith et al. 1992). For example, rejection of allogenic grafts is more slowly in larvae than in the adult individuals. Antibodies of larval amphibians express lower affinity to antigens than antibodies of the adults. It is presumed that changes in the amphibians immune system are related, on the one hand, with the larval lymphocytes decline during the metamorphosis and on the other hand, with development of new, different lymphocytes clones in the adults. Elimination of the primordial lymphocytes by programmed cell death – apoptosis, is probably mediated by the thyroid hormones, which play a crucial role in the whole metamorphosis (Ishizuya-Oka et al. 2010).

There has been an extensive study of the amphibian’s lymphoid system with the emphasis on the morphological aspect. Principal lymphoid organs counteracting on the amphibian’s lymphoid system are the **thymus, spleen, gut-associated lymphoid tissue** (GALT), and accumulations of T-lymphocytes functionally resembling the mammalian lymph nodes named **jugular bodies**. Additional “outputs” for the generation of lymphoid cells are **bone marrow** as well as liver and kidneys.

Thymus is regarded as the central lymphoid organ and the major progenitor of immune competence in amphibians. Furthermore, early thymectomy in larval *Xenopus* impairs both humoral and cell-mediated immunity (Borysenko 1976; Duellman & Trueb 1986; Lofts 2012). In amphibians, the two lobes of thymus lie near the angle of the jaws (Nelsen 1953). In larvae of *Rana catesbeiana* Shaw, 1802 (order Anura) the thy-

mus arises through an invagination of the dorsal epithelium of the second pharyngeal pouch (Du Pasquier et al. 1989). According to Curtis et al. (1972), the thymus of the post-metamorphic *Rana* is highly differentiated at the end of 2nd week. Urodele thymus lies under the skin in a close proximity of the muscles moving the gill arches. Usually three thymic primordia are situated along the each side of the body. Thymus in species of *Salamandra* lies marginally to the parotid glands and has an irregular shape, septated by connective tissue into separate groups of lobules. Unlike *Salamandra*, in *Triturus* sp. (newts) thymus is of regular uniform shape (Cooper 1982). It is generally stated that thymus in Urodela participates in generating both cellular as well as humoral immunity (Cooper et al. 1975).

Like other jawed vertebrates, the amphibian thymus consists of lymphoid cells within epithelial framework and various characteristic cell types. Amphibians' thymus comprises of epithelial reticulum enclosed by the connective tissue capsule penetrating into the parenchyma forming septa and heterogeneous microenvironment ensuring physiologically specific conditions for developing lymphocytes, thus containing cellular populations of macrophages, dendritic and numerous thymic myoid cells. Thymus parenchyma is compartmented into cortex and medulla. Even though characteristic cortico-medullary distinction and typical medullary Hassall's bodies are absent in Urodela, secretion-containing cystic formations are present in medulla. In Anura, the thymic tissue shows strictly separated cortex with densely packed lymphocytes and centrally localized medulla, in which the epithelial component is more conspicuous. Derivates of epithelial component include unicellular and multicellular Hassall's bodies, cysts and cells with eosinophilic inclusions. Anuran thymus undergoes specific **season-dependent morphological changes**, following the outer environment temperature. During the warm months, thymic tissue is clearly differentiated into the cortical and medullary regions, whereas the cold months cause decrease in cortex cellular portion of lymphocytes with subsequent increase in amount of cystic medullary formations that arise as a consequence of epithelial cells degenerative transformation (Manning & Horton 1969, 1982; Tochinnai 1976). In Urodela, the thymus does not show a distinct differentiation into typical cortical and medullary zones. In contrast with other terrestrial vertebrates, urodele thymus contains a homogeneous cortical-like tissue spotted by medulla-like areas. From the cytological point of view, *Pleurodeles* thymus does not differ from the anuran and higher vertebrate thymic ultra-architecture. During metamorphosis no more epithelial and lymphoid differentiation occurs but an increase in the number of phagocytosing cells and some new cellular populations can nevertheless be observed, e.g. mast cells, myoid cells and rare plasma cells (Manning 1979; Henry & Charlemagne 1980).

In general, anuran amphibians possess better organized **gut-associated lymphoid tissue** while urodele amphibians do not (Goldstine et al. 1975). The struc-

ture of the **spleen** in amphibians is similar to that of fish, with the exception of reticular cells presentation in the follicle areas (Zapata & Amemiya 2000).

Bone marrow firstly emerged in adult Anura and Urodela, representing the privileged place of lymphopoiesis and myelopoiesis. Gymnophiona lack the bone marrow and process of lymphopoiesis is performed in liver, kidneys, thymus and spleen. In embryonic period, the anuran kidneys have predominantly erythropoietic function. During metamorphosis, spleen of amphibians undergoes a temporary atrophy with subsequent turnover of original hematopoietic function. In further ontogenesis spleen serves as a home place for B lymphocytes maturation as well as a site of concrete immune reaction. The **univentricular lymphatic hearts** are characteristic organs for Amphibia with typical localization between lymphatic, arteries and veins anastomoses (Liu & Casley-Smith 1989). They ensure unidirectional lymph flow progressing into the blood circulation. The number of lymphatic hearts varies. Anurans possess approximately 4 to 6 lymphatic hearts in jugular, lumbar and caudal body region, whereas in Gymnophiona their amount may reach up to 200. Lymph movement is accomplished primarily by specialized lymph muscles and lung ventilation that move lymph through highly compartmentalized lymph sacs to the dorsally located lymph hearts, which are responsible for pumping lymph into the circulatory system; however, it is unclear how lymph reaches the lymph heart (Hedrick et al. 2014). Lymphatic nodes are absent in Amphibia, but some frogs (e.g., *Rana pipiens* Schreber, 1782) possess functionally resembling **jugular bodies**. The jugular bodies are surrounded by a capsule and are plasma cells and antibody-forming organs (Zapata et al. 1981a, b).

Thymus and other lymphoid organs in reptiles

It is important to take into consideration that reptiles have not been studied so extensively like amphibian. Reptile lymphoid cells and tissues are widely distributed throughout the organ systems. The lymphoid complex of turtles and lizards shows well-organized thymus, spleen, small and primitive lymph nodes and intestinal and cloacal lymphoid aggregates (Borysenko & Cooper 1972). Lymphoid aggregates are found also in lungs, kidneys, urinary bladder and along the vascular system in some species but are transient rather than permanent structures (Johnston 1973; Muthukkaruppan et al. 1982).

From the embryological point of view the thymus development in reptiles is really variable, considering the definite anatomical location of mature thymus and the lobar pattern. Thymic primordia are dorsal outgrowths of the pharyngeal pouches. Wide variability of definite thymus position in lizards, snakes and turtles develop from different pairs of pharyngeal pouches. In the lizards, the thymus arise from the second and third, in snakes thymus primordial outgrows from the fourth and fifth, in turtles thymic primordia originate in

third and fourth pharyngeal pouches (Bockman 1970; Muthukkaruppan et al. 1982; Vasse 1983).

The reptilian thymus is usually lobulated and the number of lobes is variable. In most of the lizards and snakes, there are two lobes of thymic tissue located on each side of the cervical region. On periphery cortical and centrally accumulated medullary compartments are clearly distinguished but with no further lobules subdivision. Crocodilian thymus resembles the birds' thymus the most. It is comprised of elongated, chain-like structure, passing from the skull base along the cervical region, extending nearly to the heart. Thymus in turtles presents a unique structure in that the thymus represents only a single lobe on each side of the neck, near the common carotid arteries bifurcation. Each thymic lobe is subdivided into the partial lobules with the connective tissue septa. Cortex usually comprises of small densely packed lymphocytes trapped in stellate thymic-epithelial framework. In medulla, the situation is different – only fewer lymphocytes can be found and the majority of the cellular medullary population is formed by thymic-epithelial cells. Macrophages as well as the myoid cells also participate on the thymic microenvironment formation. Reptilia lack the typical concentric Hassall's bodies. The medullary compartment of the thymus is characterized by cystic formations with centrally localized cells in the heterogeneous stage of degeneration. Involution of the primary lymphoid organ follows the very same pattern as in the human. Age-related changes progress in the manner of replacing the active thymic parenchyma with the connective tissue elements (Bockman 1970; Muthukkaruppan et al. 1982; Sypek & Borysenko 1988).

The most-developed peripheral lymphoid organ of large immunological importance is the spleen. **The spleen** of reptilian exhibits a somewhat more elaborate microanatomical structure, but still lacks the germinal centers that were first recognized in the spleen of birds (Bao et al. 2009). In the most reptilians, the spleen has spherical or elongated shape with its location adjacent to the transverse colon. It is encapsulated by a fibrous connective tissue with the processes invaginating into the parenchyma forming typical trabeculae. The parenchyma of the spleen shows a definite demarcation into red and white pulp in the most species (Tischendorf 1985; Sypek & Borysenko 1988).

The lymph nodes (in reptiles in **primitive form**) represent the only major recent innovation of the adaptive immune systems in vertebrates. They provide highly sophisticated tissue environments that promote the cellular interactions that lead to T lymphocytes activation and the affinity maturation of antibodies, and consequently, efficient memory responses (Junt et al. 2008; Mueller & Germain 2009). The crucial evolutionary transition appears to have occurred with the emergence of reptiles and birds, because some of these species are known to have a primordial form of lymph nodes and Peyer's patches. Primitive lymph nodes in reptiles are both small and poorly developed or absent in some species. In turtles are most promi-

nent as a lymphoid aggregates located in the axillary and inguinal regions (Borysenko & Cooper 1972; Borysenko 1976). Although lymphoid aggregates are commonly found also in the cloacal region, there is still no sign of forming a true avian bursa of Fabricius phylogenetic predecessor (Borysenko & Cooper 1972). In reptilian, the absence of nodules with germinal centers has been reported. This apparent lack may be a characteristic feature of all ectothermic vertebrates. However, serological studies proved that the reptilians produce secondary responses that are either quantitatively higher or/and qualitatively differ from the primary immune responses. Despite of evident lack of nodules with germinal centers in reptiles, which are associated with the secondary humoral responsiveness and immunological memory in mammals, such reactions in cold-blooded vertebrates obviously occur in the absence of mentioned structures in peripheral lymphoid organs (Wetherall & Turner 1972; Borysenko 1976).

Thymus and other lymphoid organs in birds

Most of the research on the avian immune system has been performed on the chicken. The avian immune system differs from that of mammals, mainly in the presence or absence of lymphoid organs and in the histology of these organs, in the antibody response, in the generation of diversity of antibodies and in the transfer of maternal immunity (Koch 1991; Sharma 1997). The avian lymphoid system constitutes an anatomically diffused but strategically distributed defense network against infectious organisms and other injurious elements. According to their function and role in development, they are classified into primary (central) organs that produce lymphoid precursor cells and secondary (peripheral) lymphoid organs that gather the mature immunocytes and serve as a site of immune responsiveness. The primary lymphoid organs of poultry are the embryonic yolk sac, thymus, bursa of Fabricius and bone marrow. Secondary lymphoid organs include the spleen, caecal tonsils and Harderian gland in orbito-nasal region. Following the avian maturation process the distinction between central and peripheral lymphatic organs becomes less obvious (Seto 1981).

Thymus is a paired organ comprising of about 14 lobes, 7 on each side of cervical region, adjacent to the jugular vein. In birds, several thymic lobes are arranged in a string-like pattern throughout the neck (Kendall 1980; Boehm et al. 2012a). Avian thymus can be histologically divided into cortex and medulla. These two different compartments are of very similar cellular composition as the mammalian thymus, thus expressing presence of lymphocytes, epithelial cells forming an ultrastructural network, macrophages, dendritic cells and fibroblasts. Thymic microenvironment sets appropriate conditions for T lymphocytes maturation and differentiation processes. Information about avian thymus involution is rare but regularly includes the cell density changes; however these transformations show extensive variations among the birds' species. Thymic tissue of

Gallus domesticus (L., 1758) shows obvious age-related changes including loss of cortico-medullary distinction. On the other hand, in *Columba livia* Gmelin, 1789, no apparent age-dependent changes were encountered (Seto 1981; Arstilla et al. 1994; Ciriaco et al. 2003).

Bursa of Fabricius (*bursa cloacalis Fabricii*) is organ peculiar and unique to the birds. The history of research of avian bursa cloacalis began with its description by the Italian anatomist and “The Father of Embryology” Hieronymus Fabricius (1533–1619) and with the discovery of its pivotal role in humoral immunity in the 1950s (Dorko et al. 2013). A hollow diverticulum of a proctodeal region consisting of blind, crypt-like invaginations (plicae) is located dorsally to the cloaca, is of round or oval shape in the chicken, elongated in the duck. Bursa contains 10 to 15 folds lining its lumen, having 8,000 to 12,000 total follicles. As a central lymphoid organ bursa is responsible for B lymphocytes lymphopoiesis and lymphocyte maturation, however B lymphocytes development begins in the spleen. Antibody diversity is generated in this organ. As a primary lymphoid organ the bursa of Fabricius is at hatch in precocial birds, reaches its maximum size from fifth to twelfth week of age. Bursal tissue involutive changes are variable among species. Involution depends on the increasing age, sexual maturation or on the onset of egg production (Glick et al. 1956; Bockman & Cooper 1973; Glick 1979; Kendall 1980; Lillehoj & Trout 1996; Ciriaco et al. 2003; Scott 2004).

In *Aves* as well as in mammals, the **mucosa-associated lymphoid tissue** (MALT) has been evolved as a consequence of permanent exposure of the gastrointestinal tract to the foreign environmental antigens. MALT possesses specialized features that reflect their role as a first line of defense on mucosal surfaces. Gut-associated lymphoid tissue (GALT) represents a component of MALT. GALT includes organized lymphoid structures such as **caecal tonsils**, **Peyer's patches**, **Meckel's diverticulum** and lymphocytes aggregates scattered inside the mucosa of the digestive tube (Oláh et al. 1984; Lillehoj & Trout 1996).

In some ways, the avian **spleen** is simpler than the mammalian, since it is not described to have significant erythrocytes storage function and its role in erythropoiesis is restricted. In mature birds the spleen contributes to immune defense in two ways. Firstly, by blood filtration, including phagocytosis of damaged cells and antigens. Secondly, with the production, maturation and storage of lymphocytes those are responsible for humoral and cellular immune responses. Distinctive structural feature is the predominance of ellipsoids, forming a compartment separate from the T lymphocytes rich periarteriolar lymphoid sheath. A major difference in the onset of humoral defense responses is the avian spleen capacity to establish germinal centers. Canonical germinal centers were firstly recognized in the *Aves*. Spleen, in birds, has still dominant position in generating immune responses, since the lymph nodes are still absent. Although postnatally tonsils in birds can develop along the gastrointestinal tract (John

1994; Smith & Hunt 2004; Bao et al. 2009; Balogh & Lábadi 2011).

Kondo (1937) and Biggs (1957) detected particular mural lymphoreticular formations in the walls of avian lymphatic vessels. These mural aggregates seem to be identical with the primitive lymph-node primordia in reptilians (Muthukkaruppan et al. 1982). Jolly (1913) concluded that **lymph nodes** are present only in swamp, sea and shore birds. Further histological studies mention universal pattern concerning the highly differentiated avian lymph nodes localized in thoracic and lumbar region that had been described in ducks, geese and swans. These thoracic and lumbar lymph nodes are characteristic by an irregular system of lymph sinuses and lymphoreticular tissue “cords”. The structure of the avian lymph node is characterized by a thin capsule, thin lymphoreticular cords and absence of trabeculae. There is no distinct cortex, paracortex and medulla, or subdivision of sinuses system into marginal, trabecular and medullar portions. The lymphoreticular cords in birds possess germinal centers with B lymphocytes and area of T lymphocytes (Berens von Rautenfeld & Budras 1983). Similarly to the lymph nodes, the lymphoid nodules show the irregular occurrence throughout the neck, wing and hind limb (Kihara & Naito 1933). The striking difference between lymphoid system of wild-living and domestic birds is that domestic birds do not possess the lymph nodes. Instead of lymph-node formations, the lymphoid nodules are detectable in the avian bone marrow. Although it is not clear how far this statement can be applied to all aquatic species (White 1976).

The lymph heart is a sac-like structure on either side of avian tail. In the chick, the lymph hearts are functional only *in ovo*, and after hatching they partially degenerate. In some birds, e.g. duck and emu, the lymph assists copulatory organ erection of male adults and lymph hearts function postnatally to return lymph from the lymphatic erectile phallus to the venous system (Budras et al. 1987; Valasek et al. 2007).

Harderian gland (gland of Harder) was firstly described in 1694 as *glandula nova lacrymalis*. Gland is located in orbit and from histological view it is a compound tubular or a compound tuboalveolar gland. One of the main characteristics of the gland is the extreme variety not only in its morphology, but also in its biochemical properties. Proposed functions are remarkably diverse including a gland being a site of immune response, a site of osmoregulation and a source of growth factor etc. (Payne 1994; Chieffi et al. 1996).

Thymus and other lymphoid organs in mammals

Majority of representatives of the mammalian classes possess the identical lymphatic organs as a human. Embryologically, the mammalian thymus is derived from the endodermal layer of pharyngeal part of primitive gut with important contribution of the neural crest derived cells (Grapin-Botton & Constam 2007). The third pair of pharyngeal pouches each forms a common

thymus-parathyroid primordium that will later develop into one thymus lobe and one parathyroid gland (Varga et al. 2011c).

Relatively striking interspecies differences are observed in morphology of the **mammalian thymus**, considering the variability in thoracic and/or cervical portion presentation of the gland (Boehm et al. 2012a). Diprotodont marsupial species, including kangaroo, wallabies, and possums, have both cervical, being a dominant, and thoracic thymuses (Wong et al. 2011). In sheep, cattle, goat, pigs, and horse distinct cervical thymuses are present (Jordan 1976). In particular breeds of mice, the incidence of small cervical thymus has been proved (Terszowski et al. 2006; Dooley et al. 2006). In humans, however, the thymus in the neck is a rare congenital anomaly caused by disrupted embryonic migration of the thymus to its target position (Rodewald 2008; Varga et al. 2011a). Postnatally the mammalian thymus undergoes extensive age-related involution characterized by reduction of size, cellularity of thymic tissue and lymphopoietic function.

While, in the mammals, a site of T lymphocytes maturation is exclusively in the thymus, a place for B lymphocytes maturation is variable in some species. Mostly, B lymphocytes undergo differentiation process in the **bone marrow**, however in mice B lymphocytes complete their development in the spleen, in case of bovine in the lymphatic nodes and in goats, rabbits and pork in the specific Peyer's patches of the small intestine.

Depending on the mammals species, the development of **secondary lymphoid organs** is completed during embryonic development (in humans) or shortly after birth (in mice), resulting in highly compartmentalized structure with the clustered arrangement of hemopoietic and mesenchymal cells. However, substantial rearrangement of the original resting tissue architecture occurs as a consequence of exposure to external antigens (Balogh 2011). Recent evidence has shed light also on a previously unknown role of the spleen in the development of specific B lymphocyte populations that are involved in the initial response to infection caused by encapsulated bacteria (Brendolan et al. 2007). Mammalian **spleen** is a highly vascularized organ that filters the damaged or old cells and foreign particles on one hand (splenic red pulp), and forms a secondary lymphoid organ in other hand (white pulp). In the spleen of mammalian, a specialized antigen-trapping device has emerged – the cellular microenvironment surrounding arterioles, populated by a particular type of macrophages and dendritic cells (Cesta 2006; Boehm et al. 2012a).

In mammals, **lymph nodes** are universally established as key elements of the adaptive immune system (Boehm et al. 2012a). The canonical lymph node consists of encapsulated lymphoid lobules, vessels and sinuses, and acts as a local station as lymphocytes responses to infections (van de Pavert & Mebius 2010; Brinkworth & Thorn 2013). In some mammals (pork, rhinoceros, elephant and dolphin) as a structure of

lymph nodes is reversed so a lymph flow is inverse – so called **inverted lymph nodes**. In this type of lymph node, the afferent lymphatic vessels enter the node in a site of hilus. Lymph then flows through the central cortex formed by lymphatic nodules, subsequently drains out through the peripheral medulla and collects in the subcapsular sinus, leaving the node by multiple efferent vessels (Spalding & Heath 1987). In ruminants and rodents **haemal lymph nodes** (*nodi lymphohaemales*) are present, which structurally resemble combination of the spleen and the lymph node. Hemal lymph nodes are covered by a thin capsule of connective tissue. Inside, the reticular meshwork is full filled with a large number of free blood cells, macrophages, lymphocytes and plasma cells, also numerous blood sinusoids engorged with blood are visible (Zidan & Pabst 2010). From the functional point of view, they participate on the blood filtration and erythrophagocytosis (like red pulp of the spleen), perform erythropoiesis and create thrombocytes (like bone marrow) and produce the antibodies (like white pulp of the spleen and lymph nodes) (Ezeasor & Singh 1988; Cerittu & Geurrero 2008). Haemal lymph nodes are located within the blood circulation, probably have only afferent lymphatic vessels and their color is thank to phagocytosis of red blood cells reddish or brownish – these are the most prominent differences between lymph nodes and hemal lymph nodes (Castenholz & Castenholz 1996; Cerutti & Guerrero 2008).

In mammals, evolutionary novelty in form of the **tonsils** in the wall of pharynx emerged, with their microscopic structure similar to the lymph node (presence of lymphoid nodules) as well as thymus (reticulation of pharyngeal surface epithelium, absence of afferent lymph vessels) (Slipka & Slipka 1996). Lymphoid follicles of the gut (gut-associated lymphoid system) in humans and several another mammals (e.g., horse, sheep, pig, dog, mouse, guinea pig ...) form **Peyer's patches**. Peyer's patches have various dimensions and shape, they are arranged on the anti-mesenteric border of small intestine. These aggregations of lymphoid follicles are extended from the mucosa to the submucosa, and in human their number is 239 units at puberty. Each lymphoid follicle of Peyer's patch plays a fundamental role as a generating site of antibodies, and the interfollicular areas are rich in T lymphocytes and in high endothelial venules (bidirectional pathways for lymphocytes migration between the blood stream and connective tissue) (Azzali 2003). Antigens (especially intestinal microbial antigens) from the lumen of gut are delivered to Peyer's patches by specialized microfold epithelial cells (M cells) and are present to immunologically active cells (lymphocytes and macrophages). The germinal centers of lymphoid follicles not only contribute to the generation of B lymphocytes and plasma cells producing somatically mutated gut antigen-specific IgA antibodies but have also been suggested to support non-specific antigen diversification of the B lymphocytes repertoire (Reboldi & Cyster 2016; Ohno 2016). In addition, studies in sheep and rabbits have led to the concept that the lower hindgut gut-associated lymphoid tissue rep-

resents primary lymphoid tissue for B lymphocytes and is necessary for normal B lymphocytes development, analogous to the bursa of Fabricius in birds (Butler & Sinkora 2013).

References

- Adamkov M., Furljová M., Horáček J., Benčat M. & Kružliak P. 2014. Relationship of mismatch repair proteins and survivin in colon polyps and carcinomas. *Acta Histochem.* **116** (6): 1007–1014. DOI: 10.1016/j.acthis.2014.04.005
- Adamkov M., Halasova E., Rajcani J., Bencat M., Vybohova D., Rybarova S. & Galbavy S. 2011. Relation between expression pattern of p53 and survivin in cutaneous basal cell carcinomas. *Med. Sci. Monit.* **17** (3): BR74–80. DOI: 10.12659/MSM.881442
- Agius C. & Roberts R. J. 2003. Melano-macrophage centres and their role in fish pathology. *J. Fish Dis.* **26** (9): 499–509. DOI: 10.1046/j.1365-2761.2003.00485.x
- Aifantis I., Mandal M., Sawai K., Ferrando A. & Vilimas T. 2006. Regulation of T-cell progenitor survival and cell-cycle entry by the pre-T-cell receptor. *Immunol. Rev.* **209** (1): 159–169. DOI: 10.1111/j.0105-2896.2006.00343.x
- Alitheen N.B., McClure S. & McCullagh P. 2010. B-cell development: one problem, multiple solutions. *Immunol. Cell. Biol.* **88** (4): 445–450. DOI: 10.1038/icb.2009.119
- Amemiya C.T., Saha N.R. & Zapata A. 2007. Evolution and development of immunological structures in the lamprey. *Curr. Opin. Immunol.* **19** (5): 535–541. DOI: 10.1016/j.coi.2007.08.003
- Ardavin C.F., Gomariz R.P., Barrutia M.G., Fonfria J. & Zapata A. 1984. The lympho-hemopoietic organs of the anadromous sea lamprey, *Petromyzon marinus*. A comparative study throughout its life span. *Acta Zool. (Stockh.)* **65** (1): 1–15. DOI: 10.1111/j.1463-6395.1984.tb00805.x
- Ardavin C.F. & Zapata A. 1987. Ultrastructure and changes during metamorphosis of the lympho-hemopoietic tissue of the larval anadromous sea lamprey *Petromyzon marinus*. *Dev. Comp. Immunol.* **11** (1): 79–93.
- Arstilla T.P., Vainio O. & Lassila O. 1994. Central role of CD4+ T cells in avian immune response. *Poultry Sci.* **73** (7): 1019–1026. DOI: 10.3382/ps.0731019
- Austbø L., Aas I.B., König M., Weli S.C., Syed M., Falk K. & Koppang E.O. 2014. Transcriptional response of immune genes in gills and the interbranchial lymphoid tissue of Atlantic salmon challenged with infectious salmon anaemia virus. *Dev. Comp. Immunol.* **45** (1): 107–114. DOI: 10.1016/j.dci.2014.02.007
- Aw D. & Palmer D.B. 2011. The origin and implication of thymic involution. *Aging Dis.* **2** (5): 437–443. PMID: 22396892
- Azzali G. 2003. Structure, lymphatic vascularization and lymphocyte migration in mucosa-associated lymphoid tissue. *Immunol. Rev.* **195** (1): 178–193. DOI: 10.1034/j.1600-065X.2003.00072.x
- Bai M., Doukas M., Papoudou-Bai A., Barbouti A., Stefanaki K., Galani V. & Kanavaros P. 2013. Immunohistological analysis of cell cycle and apoptosis regulators in thymus. *Ann. Anat.* **195** (2): 159–165. DOI: 10.1016/j.aanat.2012.07.012
- Bajoghli B., Guo P., Aghaallaei N., Hirano M., Strohmeier C., McCurley N., Bockman D.E., Schorpp M., Cooper M.D. & Boehm T. 2011. A thymus candidate in lampreys. *Nature* **470** (7332): 90–94. DOI: 10.1038/nature09655.
- Balogh P. 2011. Introduction: Evolution of peripheral lymphoid organs, Chapter 1, pp. 1–3. DOI: 10.1007/978-3-642-14429-5_1 In: Balogh P. (ed.), *Developmental Biology of Peripheral Lymphoid Organs*, Springer-Verlag, Berlin, Heidelberg, 177 pp. ISBN: 978-3-642-14428-8
- Balogh P. & Lábadi, A. 2011. Structural evolution of the spleen in man and mouse, Chapter 11, pp. 121–141. DOI: 10.1007/978-3-642-14429-5_11 In: Balogh P. (ed.), *Developmental Biology of Peripheral Lymphoid Organs*, Springer-Verlag, Berlin, Heidelberg, 177 pp. ISBN: 978-3-642-14428-8
- Bao H.J., Li M.Y., Wang J., Qin J.H., Xu C.S., Hei N.N., Yang P., Gandahi J.A. & Chen Q.S. 2009. Architecture of the blood-spleen barrier in the soft-shelled turtle, *Pelodiscus sinensis*. *Anat. Rec. (Hoboken)* **292** (8): 1079–1087. DOI: 10.1002/ar.20917.
- Beard J. 1894. The development and probable function of the thymus. *Anat. Anz.* **9**: 476–486.
- Berens von Rautenfeld D. & Budras K.D. 1983. Topography, ultrastructure and phagocytic capacity of avian lymph nodes. *Cell Tissue Res.* **228** (2): 389–403. DOI: 10.1007/BF00204887
- Biggs P.M. 1957. The association of lymphoid tissue with the lymph vessels in the domestic chicken (*Gallus domesticus*). *Acta Anat.* **29** (1–2): 36–47. DOI: 10.1159/000141159
- Bockman D.G. 1970. The thymus, Chapter 4, pp. 111–130. In: Gans C. & Parsons T.S. (eds), *Biology of the Reptilia*, Vol. 3, Part C Morphology, Academic Press, New York, 385 pp.
- Bockman D.E. & Cooper M.D. 1973. Pinocytosis by epithelium associated with lymphoid follicles in the bursa of Fabricius, appendix and Peyer's patches. An electron microscopic study. *Am. J. Anat.* **136** (4): 455–477. DOI: 10.1002/aja.1001360406
- Bodey B., Bodey B.Jr., Siegel S.E. & Kaiser H.E. 2000. Novel insights into the function of the thymic Hassall's bodies. In *Vivo* (Athens, Greece) **14** (3): 407–418. PMID: 10904874
- Boehm T. 2011. Design principles of adaptive immune system. *Nat. Rev. Immunol.* **11** (5): 307–317. DOI: 10.1038/nri2944.
- Boehm T. & Bleul C.C. 2007. The evolutionary history of lymphoid organs. *Nat. Immunol.* **8** (2): 131–135. DOI: 10.1038/ni1435
- Boehm T., Hess I. & Swann J.B. 2012a. Evolution of lymphoid tissues. *Trends Immunol.* **33** (6): 315–321. DOI: 10.1016/j.it.2012.02.005.
- Boehm T., McCurley N., Sutoh Y., Schorpp M., Kasahara M. & Cooper M.D. 2012b. VLR-based adaptive immunity. *Annu. Rev. Immunol.* **30**: 203–220. DOI: 10.1146/annurev-immunol-020711-075038
- Borysenko M. 1976. Phylogeny of immunity: An overview. *Immunogenetics* **3** (1): 305–326. DOI: 10.1007/BF01576964
- Borysenko M. & Cooper E.L. 1972. Lymphoid tissue in the snapping turtle, *Chelydra serpentina*. *J. Morphol.* **138** (4): 487–497. DOI: 10.1002/jmor.1051380408
- Bowden T.J., Cook P. & Rombout J.H. 2005. Development and function of the thymus in teleosts. *Fish Shellfish Immunol.* **19** (5): 413–427. DOI: 10.1016/j.fsi.2005.02.003
- Brendolan A., Rosado M.M., Carsetti R., Salleri L. & Dear T.N. 2007. Development and function of the mammalian spleen. *Bioessays* **29** (2): 166–177. DOI: 10.1002/bies.20528
- Bringworth J.F. & Thorn M. 2013. Vertebrate immune system evolution and comparative primate immunity, pp. 17–64. DOI: 10.1007/978-1-4614-7181-3_2. In: Bringworth J.F. & Pechenkina K. (eds), *Primates, Pathogens, and Evolution*, Part I, series *Developments in Primatology: Progress and Prospects*, Springer, New York, Heidelberg, Dordrecht, London. 428 pp. ISBN: 978-1-4614-7180-6
- Budras K.-D., Hullinger R.L. & Rautenfeld D.B.V. 1987. Lymph heart musculature in birds. *J. Morphol.* **191** (1): 77–87. DOI: 10.1002/jmor.1051910108
- Burne R.H. 1926. Anatomy of the ductless glands and lymphatic system of the angler fish (*Lophius piscatorius*). *Philos. Trans. R. Soc. London* **215** (421–430): 1–56. DOI: 10.1098/rstb.1927.0001
- Butler J.E. & Sinkora M. 2013. The enigma of the lower gut-associated lymphoid tissue (GALT). *J. Leukoc. Biol.* **94** (2): 259–270. DOI: 10.1189/jlb.0313120
- Butler J.E., Sinkora M., Wertz N., Holtmeier W. & Lemke C.D. 2006. Development of the neonatal B and T cell repertoire in swine: implications for comparative and veterinary immunology. *Vet. Res.* **37** (3): 417–441. DOI: 10.1051/vetres:2006009
- Castenholz A. & Castenholz H.E. 1996. Casting methods of scanning electron microscopy applied to hemal lymph nodes in rats. *Lymphology* **29** (3): 95–105.
- Cerutti P. & Guerrero F. 2008. Erythrophagocytosis and erythrophagocytosis in bovine haemal nodes. *Int. J. Morphol.* **26** (3): 557–562. DOI: 10.4067/S0717-95022008000300008

- Cesta M.F. 2006. Normal structure, function and histology of the spleen. *Toxicol. Pathol.* **34** (5): 455–465. DOI: 10.1080/01926230600867743
- Chantanachookhin C., Seikai T. & Tanaka M. 1991. Comparative study of the ontogeny of the lymphoid organs in three species of marine fish. *Aquaculture* **99** (1–2): 143–155. DOI: 10.1016/0044-8486(91)90294-H
- Chieffi G., Chieffi Bacari G., Di Matteo L., d'Istria M., Minucci S. & Varriale B. 1996. Cell biology of the Harderian gland. *Int. Rev. Cytol.* **168**: 1–80. DOI: 10.1016/S0074-7696(08)60882-7
- Chilmonczyk K. 1992. The thymus in fish: Development and possible function in the immune response. *Annu. Rev. Fish Disease* **2**: 181–200. DOI: 10.1016/0959-8030(92)90063-4
- Chung C.Y., Ysebaert D., Berneman Z.N. & Cools N. 2013. Dendritic cells: cellular mediators for immunological tolerance. *Clin. Dev. Immunol.* **2013**: 8 pp. DOI: 10.1155/2013/972865.
- Ciriaco E., Pinera P.P., Diaz-Esnal B. & Laura R. 2003. Age-related changes in the avian primary lymphoid organs (thymus and bursa of Fabricius). *Microsc. Res. Tech.* **62** (6): 482–487. DOI: 10.1002/jemt.10416
- Cooper E.L. 1982. *General Immunology*. A Wheaton & Co. Ltd., Exeter, 343 pp. ISBN: 0080263682, 9780080263687
- Cooper M.D. & Alder, M.N. 2006. The evolution of adaptive immune systems. *Cell* **124** (4): 815–822. DOI: 10.1016/j.cell.2006.02.001
- Cooper E.L., Brown B.A. & Wright R.K. 1975. New ideas on amphibian immunity: the lymph gland: a generator of both T and B cells. *Am. Zool.* **15** (1): 85–92. DOI: 10.1093/icb/15.1.85
- Curtis S.K., Volpe E.P. & Cowden R.R. 1972. Ultrastructure of the developing thymus of the leopard frog. *Rana pipiens*. *Z. Zellforsch. Mikrosk. Anat.* **127** (3): 323–346. DOI: 10.1007/BF00306877
- Dalum A.S., Austbø L., Bjørgen H., Skjødt K., Hordvik I., Hansen T., Fjelldal P.G., Press C.M., Griffiths D.J. & Koppang E.O. 2015. The interbranchial lymphoid tissue of Atlantic Salmon (*Salmo salar* L) extends as a diffuse mucosal lymphoid tissue throughout the trailing edge of the gill filament. *J. Morphol.* **276** (9): 1075–1088. DOI: 10.1002/jmor.20403
- Danilova N. 2006. The evolution of immune mechanisms. *J. Exp. Zool. B. Mol. Dev. Evol.* **306B** (6): 496–520. DOI: 10.1002/jez.b.21102
- Danilova N., Hohman V.S., Sacher F., Ota T., Willett C.E. & Steiner L.A. 2004. T cells and the thymus in developing zebrafish. *Dev. Comp. Immunol.* **28** (7–8): 755–767. DOI: 10.1016/j.dci.2003.12.003
- Dooley J., Erickson M., Gillard G.O. & Farr A.G. 2006. Cervical thymus in the mouse. *J. Immunol.* **176** (11): 6484–6490. DOI: 10.4049/jimmunol.176.11.6484
- Dorko F., Danko J., Flešárová S., Boroš E. & Sobeková A. 2011a. Effect of pesticide bendiocarbamate on distribution of acetylcholine- and butyrylcholine-positive nerves in rabbit's thymus. *Eur. J. Histochem.* **55** (4): e37. DOI: 10.4081/ejh.2011.e37
- Dorko F., Horáček J., Tokarčík J. & Miko M. 2013. Cholinesterase activity in quail primary lymphoid organs. *Biologia* **68** (6): 1238–1242. DOI: 10.2478/s11756-013-0269-0
- Dorko F., Kluchová D., Boleková A., Špakovská T., Borošová T. & Lovasová K. 2011b. Influence of surgical and chemical orchidectomy on weight and distribution of AChE-nerve fibers in thymuses of adult rats. *Eur. J. Histochem.* **55** (3): e22. DOI: 10.4081/ejh.2011.e22
- Du Pasquier L. 1982. Ontogeny of immunological functions in amphibians, Chapter 18, pp. 633–657. DOI: 10.1007/978-1-4684-4166-6_18. In: Cohen N. & Sigel M.M. (eds), *Phylogeny and Ontogeny*, Springer US, 757 pp. ISBN: 978-1-4684-4168-0
- Du Pasquier L., Schwager J. & Flajnik M.F. 1989. The immune system of *Xenopus*. *Annu. Rev. Immunol.* **7**: 251–275. DOI: 10.1146/annurev.iy.07.040189.001343
- Duellman W.E. & Trueb L. 1986. *Biology of Amphibians*. The Johns Hopkins University Press, Maryland, 670 pp. ISBN: 080184780X
- Ekman A., Pessa-Morikawa T., Liljavirta J., Niku M. & Iivanainen A. 2010. B-cell development in bovine fetuses proceeds via a pre-B like cell in bone marrow and lymph nodes. *Dev. Comp. Immunol.* **34** (8): 896–903. DOI: 10.1016/j.dci.2010.03.012
- Elmore S. 2007. Apoptosis. A review of programmed cell death. *Toxicol. Pathol.* **35** (4): 495–516. DOI: 10.1080/01926230701320337
- Ezeasor D.N. & Singh A. 1988. Histology of the caprine hemal node. *Acta Anat.* **133** (1): 16–23. DOI: 10.1159/000146607
- Fänge R. 1977. Size relations of lymphomyeloid organs in some cartilaginous fish. *Acta Zool.* **58** (3): 125–128. DOI: 10.1111/j.1463-6395.1977.tb00246.x
- Fänge R. & Mattisson A. 1981. The lymphomyeloid (hemopoietic) system of the atlantic nurse shark, *Ginglymostoma cirratum*. *Biol. Bull.* **160** (2): 240–249.
- Fänge R. & Pulsford A. 1983. Structural studies on lymphomyeloid tissues of the dogfish, *Scyliorhinus canicula* L. *Cell Tissue Res.* **230** (2): 337–351. DOI: 10.1007/BF00213808
- Fänge R. & Sundell G. 1969. Lymphomyeloid tissues, blood cells and plasma proteins in *Chimaera monstrosa* (Pisces, Holocephali). *Acta Zool.* **50** (1–2): 155–168. DOI: 10.1111/j.1463-6395.1969.tb00537.x
- Flajnik M.F. & Du Pasquier, L. 2004. Evolution of innate and adaptive immunity: can we draw a line? *Trends Immunol.* **25** (12): 640–644. DOI: 10.1016/j.it.2004.10.001
- Fujii T. 1982. Electron microscopy of the leucocytes of the typhlosole in ammocoetes, with special attention to the antibody producing cells. *J. Morphol.* **173** (1): 87–100. DOI: 10.1002/jmor.1051730108
- Geenen V. 2012a. The appearance of the thymus and the integrated evolution of adaptive immune and neuroendocrine systems. *Acta Clin. Belg.* **67** (3): 209–213. DOI: 10.1179/ACB.67.3.2062657 –
- Geenen V. 2012b. Presentation of neuroendocrine self in the thymus: a necessity for integrated evolution of the immune and neuroendocrine systems. *Ann. N. Y. Acad. Sci.* **1261** (1): 42–48. DOI: 10.1111/j.1749-6632.2012.06624.x
- Glick B. 1979. The avian immune system. *Avian Dis.* **23** (2): 282–289. DOI: 10.2307/1589557
- Glick B., Chang T.S. & Jaap R.G. 1956. The bursa of Fabricius and antibody production. *Poultry Sci.* **35** (1): 224–225. DOI: 10.3382/ps.0350224
- Goldstine S.N., Manickavel V. & Cohen N. 1975. Phylogeny of gut-associated lymphoid tissue. *Amer. Zool.* **15**: 107–118. DOI: 10.1093/icb/15.1.107
- Gorgollon P. 1983. Fine structure of the thymus in the adult cling fish *Sicyases sanguineus* (Pisces, Gobiesocidae). *J. Morphol.* **177** (1): 25–40. DOI: 10.1002/jmor.1051770103
- Grapin-Botton A. & Constam D. 2007. Evolution of the mechanisms and molecular control of endoderm formation. *Mech. Dev.* **124** (4): 253–278. DOI: 10.1016/j.mod.2007.01.001
- Guo P., Hirano M., Herrin B.R., Li J., Yu C., Sadlonova A. & Cooper M.D. 2009. Dual nature of the adaptive immune system in lampreys. *Nature* **459** (7248): 796–801. DOI: 10.1038/nature08068
- Hadeiba H. & Butcher E.C. 2013. Thymus-homing dendritic cells in central tolerance. *Eur. J. Immunol.* **43** (6): 1425–1429. DOI: 10.1002/eji.201243192
- Halašová E., Adamkov M., Matáková T., Kavcová E., Poliaček I. & Singliar A. 2010. Lung cancer incidence and survival in chromium exposed individuals with respect to expression of anti-apoptotic protein survivin and tumor suppressor p53 protein. *Eur. J. Med. Res.* **15** (Suppl. 2): 55–59. PMID: 21147621
- Hale L.P. & Markert M.L. 2004. Corticosteroids regulate epithelial cell differentiation and Hassall body formation in the human thymus. *J. Immunol.* **172** (1): 617–624. DOI: 10.4049/jimmunol.172.1.617
- Hansen J.D. & Zapata A.G. 1998. Lymphocyte development in fish and amphibians. *Immunol. Rev.* **166**: 199–220. DOI: 10.1111/j.1600-065X.1998.tb01264.x

- Hart S., Wrathmell A.B. & Harris J.E. 1986. Ontogeny of gut-associated lymphoid tissue (GALT) in the dogfish *Scyliorhinus canicula* L. Vet. Immunol. Immunopathol. **12** (1-4): 107–116. DOI: 10.1016/0165-2427(86)90115-7
- Hedrick M.S., Hansen K., Wang T., Lauridsen H., Thygesen J. & Pedersen, M. 2014. Visualising lymph movement in anuran amphibians with computed tomography. J. Exp. Biol. **217** (17): 2990–2993 DOI: 10.1242/jeb.106906
- Henry M. & Charlemagne J. 1980. Development of amphibian thymus. Morphological differentiation, multiplication, migration and lysis of thymocytes in the urodele *Pleurodeles waltlii*. J. Embryol. Exp. Morph. **57**: 219–232. PMID: 7430931
- Herrin B.R. & Cooper M.D. 2010. Alternative adaptive immunity in jawless vertebrates. J. Immunol. **185** (3): 1367–1374. DOI: 10.4049/jimmunol.0903128
- Honma Y., Okabe K. & Chiba A. 1984. Comparative histology of the Leydig and epigonal organs in some elasmobranchs. Jap. J. Ichthyol. **31** (1): 47–54.
- Ishizuya-Oka A., Hasebe T. & Shi Y.-B. 2010. Apoptosis in amphibian organs during metamorphosis. Apoptosis **15** (3): 350–364. DOI: 10.1007/s10495-009-0422-y
- Jablonska-Mestanová V., Sisovsky V., Danisovic L., Polak S. & Varga I. 2013. The normal human newborns thymus. Bratisl. Lek. Listy **114** (7): 402–408. DOI: 10.4149/BLL_2013_086
- John J.L. 1994. The avian spleen: A neglected organ. Q. Rev. Biol. **69** (3): 327–351. DOI: 10.1086/418649
- Johnston M.R.L. 1973. Perivascular lymphoid tissue associated with the axillary lymph sinus and lateral vein of the *Gehyra variegata*. J. Morphol. **139** (4): 431–438. DOI: 10.1002/jmor.1051390405
- Jolly J. 1913. Sur les organes lympho-epitheliaux. Comp. Rend. Hebdom. Seanc. Mem. Soc. Biol. **A65**, **74** (T1): 540–543.
- Jordan R.K. 1976. Development of sheep thymus in relation to utero thymectomy experiments. Eur. J. Immunol. **6** (10): 693–698. DOI: 10.1002/eji.1830061007
- Joss J.M.P. 1998. Are lungfish neotenic? Clin. Exp. Pharmacol. Physiol. **25** (9): 733–735. DOI: 10.1111/j.1440-1681.1998.tb02286.x
- Junt T., Scandella E. & Ludewig B. 2008. Form follows function: lymphoid tissue microarchitecture in antimicrobial immune defence. Nat. Rev. Immunol. **8** (10): 764–775. DOI: 10.1038/nri2414
- Kachlik D., Baca V., Bozdechova I., Cech P. & Musil V. 2008. Anatomical terminology and nomenclature: past, present and highlights. Surg. Radiol. Anat. **30** (6): 459–466. DOI: 10.1007/s00276-008-0357-y
- Kannan T.A., Ramesh G., Ushakumary S., Dhinakarraj G. & Vairamuthu S. 2015. Thymic Hassall's corpuscles in Nandanam chicken – light and electronmicroscopic perspective (*Gallus domesticus*). J. Anim. Sci. Technol. **57**: 30. DOI: 10.1186/s40781-015-0064-2
- Katagiri Ch. 1978. *Xenopus laevis* as a model for the study of immunology. Dev. Comp. Immunol. **2** (1): 5–13. DOI: 10.1016/S0145-305X(78)80020-2
- Kendall M.D. 1980. Avian thymus glands: a review. Dev. Comp. Immunol. **4**: 191–209. DOI: 10.1016/S0145-305X(80)80023-1
- Kihara T. & Naito E. 1933. Beiträge zur Anatomie des Lymphgefäßsystems der Wirbeltiere und des Menschen (Japaner). Nr. **19**. Über den Einlagerungs- und Verbreitungsmodus des lymphatischen Gewebes im Lymph-gefäßssystem der Ente. Folia Anat. Jap. **11** (5): 405–413.
- Koch G. 1991. The immune system in poultry. Tijdschr. Diergeneeskde. **116** (14): 728–734.
- Kondo M. 1937. Beiträge zur Anatomie des Lymphgefäßsystems der Wirbeltiere und des Menschen (Japaner). Nr. **24**. Die lymphatischen gebilde im lymphgefäßssystem der Hühner. Folia Anat. Jap. **15**: 309–325.
- Konkel J.E., Jin W., Abbatiello B., Grainger J.R. & Chen W. 2014. Thymocyte apoptosis drives the intrathymic generation of regulatory T cells. Proc. Natl. Acad. Sci. USA. **111** (4): E465–473. DOI: 10.1073/pnas.1320319111.
- Koppang E.O., Fischer U., Moore L., Tranulis M.A., Dijkstra J.M., Köllner B., Aune L., Jirillo E. & Hordvik I. 2010. Salmonid T cells assemble in the thymus, spleen and in novel interbranchial lymphoid tissue. J. Anat. **217** (6): 728–739. DOI: 10.1111/j.1469-7580.2010.01305.x
- Kovalska M., Kovalska L., Pavlikova M., Janickova M., Mikuskova M., Adamkov M., Kaplan P., Tatarkova Z. & Lehotsky J. 2012. Intracellular signaling MAPK pathway after cerebral ischemia-reperfusion injury. Neurochem. Res. **37** (7): 1568–1577. DOI 10.1007/s11064-012-0752-y
- Lillehoj H.S. & Trout J.M. 1996. Avian gut-associated lymphoid tissues and intestinal immune responses to *Eimeria* parasites. Clin. Microbiol. Rev. **9** (3): 349–360. PMID: 8809465
- Liu Z. & Casley-Smith J.R. 1989. The fine structure of the amphibian lymph heart. Lymphology **22** (1): 25–30. PMID: 2725054
- Lofts B. 2012. Physiology of the Amphibian. Vol. 3. Academic Press, INC, New York, 658 pp. ISBN: 0-12-455403-2
- Luer C.A., Walsh C.J., Bodine A.B., Wyffels J.T. & Scott T.R. 1995. The elasmobranch thymus: Anatomical, histological, and preliminary functional characterization. J. Exp. Zool. **273** (4): 342–354. DOI: 10.1002/jez.1402730408
- Lutton B.V. & Callard I.P. 2008. Influence of reproductive activity, sex steroids, and seasonality on epigonal organ cellular proliferation in the skate (*Leucoraja erinacea*). Gen. Comp. Endocrinol. **155** (1): 116–125. DOI: 10.1016/j.ygcen.2007.03.011
- Manley N.R., Richie E.R., Blackburn C.C., Condie B.G. & Sage J. 2011. Structure and function of the thymic microenvironment. Front. Biosci. **16** (7): 2461–2477. DOI: 10.2741/3866
- Manning M. J. 1979. Evolution of the vertebrate immune system. J. R. Soc. Med. **72** (9): 683–688. DOI: 10.1177/014107687907200911
- Manning M.J. & Horton J.D. 1969. Histogenesis of lymphoid organs in larvae of the South African clawed toad, *Xenopus laevis* (Daudin). J. Embryol. Exp. Morph. **22** (2): 265–277. PMID: 5361557
- Manning M.J. & Horton J.D. 1982. RES Structure and Function of the Amphibia, pp. 423–459. In: Cohen N. & Sigel N.N. (eds), The Reticuloepithelial System **3**. A Comprehensive Treatise. Plenum Press, New York.
- Mattisson A. & Fänge R. 1982. The cellular structure of the Leydig organ in the shark, *Etmopterus spinax* (L.). Biol. Bull. **162** (2): 182–194. DOI: 10.2307/1540813
- Mattisson A., Fänge R. & Zapata A. 1990. Histology and ultrastructure of the cranial lymphohaemopoietic tissue in *Chimaera monstrosa* (Pisces, Holocephali). Acta Zool. **71** (2): 97–106. DOI: 10.1111/j.1463-6395.1990.tb01074.x
- Mayer S. 1888. Zur lehre von der Schilddrüse und thymus bei Amphibien. Anat. Anz. **3**: 97–103.
- Mohammad M.G., Chilmoneczyk S., Birch D., Aladaileh S., Raftos D. & Joss J. 2007. Anatomy and cytology of the thymus in juvenile Australian lungfish, *Neoceratodus forsteri*. J. Anat. **211** (6): 784–797. DOI: 10.1111/j.1469-7580.2007.00814.x
- Mravec B., Ondicova K., Valaskova Z., Gidron Y. & Hulin I. 2009. Neurobiological principles in the etiopathogenesis of disease: when diseases have a head. Med. Sci. Monit. **15** (1): RA6–16. PMID: 19114982
- Mueller S.N. & Germain R.N. 2009. Stromal cell contributions to the homeostasis and functionality of the immune system. Nat. Rev. Immunol. **9** (9): 618–629. DOI: 10.1038/nri2588
- Muthukkaruppan V.R., Borysenko M. & El Ridi R. 1982. RES Structure and function of the Reptilia, Chapter 12, pp. 461–508. DOI: 10.1007/978-1-4684-4166-6_12. In: Cohen N. & Sigel N.N. (eds), The Reticuloepithelial System. A Comprehensive Treatise, Plenum Press, New York, 757 pp. ISBN: 978-1-4684-4168-0
- Nakagawa Y., Ohigashi I., Nitta T., Sakata M., Tanaka K., Murata S., Kanagawa O. & Takahama Y. 2012. Thymic nurse cells provide microenvironment for secondary T cell receptor α rearrangement in cortical thymocytes. Proc. Natl. Acad. Sci. USA. **109** (50): 20572–20577. DOI: 10.1073/pnas.1213069109
- Nakamura H. & Ayer-Le Lièvre C. 1986. Neural crest and thymic myoid cells. Curr. Top. Dev. Biol. **20**: 111–115. DOI: 10.1016/S0070-2153(08)60658-4

- Nelsen O.E. 1953. Comparative embryology of the vertebrates. McGraw-Hill Book Company, INC. New York, 982 pp. DOI: 10.5962/bhl.title.6451
- Nishio H., Matsui K., Tsuji H., Tamura A. & Suzuki K. 2001. Immunolocalization of the mitogen-activated protein kinase signaling pathway in Hassall's corpuscles of the human thymus. *Acta Histochem.* **103** (1): 89–98. DOI: 10.1078/0065-1281-00581
- Nishino M., Ashiku S.K., Kocher O.N., Thurer R.L., Boiselle P.M. & Hatabu H. 2006. The thymus: a comprehensive review. *Radiographics* **26** (2): 335–348. DOI: 10.1148/rq.262045213
- Oguri M. 1983. On the Leydig organ in the esophagus of some elasmobranchs. *Bull. Jap. Soc. Sci. Fish.* **49** (7): 989–991. DOI: <http://doi.org/10.2331/suisan.49.989>
- Ohno H. 2016. Intestinal M cells. *J. Biochem.* **159** (2): 151–160. DOI: 10.1093/jb/mvv121
- Oláh I., Glick B. & Taylor R.L. Jr. 1984. Meckel's diverticulum. II. A novel lymphoepithelial organ in the chicken. *Anat. Rec.* **208** (2): 253–263. DOI: 10.1002/ar.1092080212
- Osório J. & Rétaux S. 2008. The lamprey in evolutionary studies. *Dev. Genes Evol.* **218** (5): 221–235. DOI: 10.1007/s00427-008-0208-1
- Ottaviani E., Franchini A. & Franceschi C. 1997. Evolution of neuroendocrine thymus: studies on POMC-derived peptides, cytokines and apoptosis in higher and lower vertebrates. *J. Neuroimmunol.* **72** (1): 67–74. DOI: 10.1016/S0165-5728(96)00146-4
- Øvergård A.C., Fiksdal I.U., Nerland A.H. & Patel S. 2011. Expression of T-cell markers during Atlantic halibut (*Hippoglossus hippoglossus* L.) ontogenesis. *Dev. Comp. Immunol.* **35** (2): 203–213. DOI: 10.1016/j.dci.2010.09.009
- Paganelli R., Giovannetti A., Pierdominici M., Di Iorio A., Cianci R., Murdaca G., Puppo F. & Pandolfi P. 2008. Apoptosis in the homeostasis of the immune system and in human immune mediated disease. *Curr. Pharm. Des.* **14** (3): 253–268. DOI: 10.2174/138161208783413310
- Page M. & Rowley A.F. 1982. A morphological study of pharyngeal lymphoid accumulations in larval lampreys. *Dev. Comp. Immunol.* **Suppl 2**: 35–40.
- Panse Le R. & Berrih-Aknin S. 2005. Thymic myoid cells protect thymocytes from apoptosis and modulate their differentiation: implication of the ERK and Akt signaling pathways. *Cell Death Differ.* **12** (5): 463–472. DOI: 10.1038/sj.cdd.4401611
- Patel S., Sørhus E., Fiksdal I.U., Espedal P.G., Bergh O., Rødseth O.M., Morton H.C. & Nerland A.H. 2009. Ontogeny of lymphoid organs and development of IgM-bearing cells in Atlantic halibut (*Hippoglossus hippoglossus* L.). *Fish Shellfish Immunol.* **26** (3): 385–395. DOI: 10.1016/j.fsi.2008.11.018
- Payne A.P. 1994. The Harderian gland: a tercentennial review. *J. Anat.* **185** (Pt 1): 1–49. PMID: 7559104
- Picchietti S., Guerra L., Buonocore F., Randelli E., Fausto A.M. & Abelli L. 2009. Lymphocyte differentiation in sea bass thymus: CD4 and CD8-alpha gene expression studies. *Fish Shellfish Immunol.* **27** (1): 50–56. DOI: 10.1016/j.fsi.2009.04.003
- Pospíšilová V., Slípka J. & Černý R. 2003. The relation of the germ layers to the ecto-mesenchyme of the neural placodes during thymus development. *Trends and Perspectives of Contemporary Morphology. 40th Congress of the Czech Anatomical Society, Plzeň, 10.-12.9.2002. Plzeň Lék. Sbor.* **78** (Suppl 1): 23–28.
- Prymak T. 1902. Beiträge zur Kenntniss des feinen Baues und der Involution der Thymusdrüse bei den Teleostieren. *Anat. Anz.* **21**: 164.
- Raica M., Cîmpean A.M., Encică S. & Cornea R. 2007. Involution of the thymus: a possible diagnostic pitfall. *Rom. J. Morphol. Embryol.* **48** (2): 101–106.
- Raica M., Encică S., Motoc A., Cîmpean A.M., Scridon T. & Bârsan M. 2006. Structural heterogeneity and immunohistochemical profile of Hassall corpuscles in normal human thymus. *Ann. Anat.* **188** (4): 345–352. DOI: 10.1016/j.aanat.2006.01.012
- Ratcliffe M.J. 2002. B cell development in gut associated lymphoid tissues. *Vet. Immunol. Immunopathol.* **87** (3–4): 337–440. PMID: 12072255
- Ratcliffe M.J. 2006. Antibodies, immunoglobulin genes and the bursa of Fabricius in chicken B cell development. *Dev. Comp. Immunol.* **30** (1–2): 101–118. DOI: 10.1016/j.dci.2005.06.018
- Reboldi A. & Cyster J.G. 2016. Peyer's patches: organizing B-cell responses at the intestinal frontier. *Immunol. Rev.* **271**(1): 230–245 DOI: 10.1111/imr.12400.
- Reyes García M.G. & García Tamayo F. 2013. The importance of the nurse cells and regulatory cells in the control of T lymphocyte responses. *BioMed. Res. Int.* **2013**: 15 pp. DOI: 10.1155/2013/352414
- Rezzani R., Bonomini F. & Rodella L.F. 2008. Histochemical and molecular overview of the thymus as site for T-cells development. *Prog. Histochem. Cytochem.* **43** (2): 73–120. DOI: 10.1016/j.proghi.2008.03.001
- Rezzani R., Nardo L., Favero G., Peroni M. & Rodella L.F. 2014. Thymus and aging: morphological, radiological, and functional overview. *Age (Dordr.)*. **36** (1): 313–351. DOI: 10.1007/s11357-013-9564-5
- Rodewald H. R. 2008. Thymus organogenesis. *Annu. Rev. Immunol.* **26**: 355–388. DOI: 10.1146/annurev.immunol.26.021607.090408
- Rodríguez R.M., López-Vázquez A. & López-Larrea C. 2012. Immune systems evolution. *Adv. Exp. Med. Biol.* **739**: 237–251. DOI: 10.1007/978-1-4614-1704-0_15.
- Rollins-Smith L.A., Blair P.J. & Davis A.T. 1992. Thymus ontogeny in frogs: T-cell renewal at metamorphosis. *Dev. Immunol.* **2** (3): 207–213. DOI: 10.1155/1992/26251
- Romano N., Fanelli M., Maria Del Papa G., Scapigliati G. & Mastrolia L. 1999a. Histological and cytological studies on the developing thymus of sharpnout seabream, *Diplodus puntazzo*. *J. Anat.* **194** (Pt 1): 39–50. DOI: 10.1046/j.1469-7580.1999.19410039.x
- Romano N., Taverne-Thiele A.J., Fanelli M., Baldassini M.R., Abelli L., Mastrolia L., Van Muiswinkel W.B. & Rombout, J.H. 1999b. Ontogeny of the thymus in a teleost fish, *Cyprinus carpio* L.: developing thymocytes in the epithelial microenvironment. *Dev. Comp. Immunol.* **23** (2): 123–137. DOI: 10.1016/S0145-305X(98)00053-6
- Rosenstiel P., Philipp E.E., Schreiber S. & Bosch T.C. 2009. Evolution and function of innate immune receptors – insights from marine invertebrates. *J. Innate. Immun.* **1** (4): 291–300. DOI: 10.1159/000211193.
- Rumfelt L.L., McKinney E.C., Taylor E. & Flajnik M.F. 2002. The development of primary and secondary lymphoid tissues in the nurse shark *Ginglymostoma cirratum*: B-cell zones precede dendritic cell immigration and T-cell zone formation during ontogeny of the spleen. *Scand. J. Immunol.* **56** (2): 130–148. DOI: 10.1046/j.1365-3083.2002.01116.x
- Saha N.R., Smith J. & Amemiya C.T. 2010. Evolution of adaptive immune recognition in jawless vertebrates. *Semin. Immunol.* **22** (1): 25–33. DOI: 10.1016/j.smim.2009.12.002
- Sarang Z., Garabuczi É., Joós G., Kiss B., Tóth K., Rühl R. & Szondy Z. 2013. Macrophages engulfing apoptotic thymocytes produce retinoids to promote selection, differentiation, removal and replacement of double positive thymocytes. *Immunobiol.* **218** (11): 1354–1360. DOI: 10.1016/j.imbio.2013.06.009
- Scott T.R. 2004. Our current understanding of humoral immunity of poultry. *Poultry Sci.* **83** (4): 574–579. DOI: 10.1093/ps/83.4.574
- Schröder M.B., Villena A.J. & Jørgensen T.O. 1998. Ontogeny of lymphoid organs and immunoglobulin producing cells in Atlantic cod (*Gadus morhua* L.). *Dev. Comp. Immunol.* **22** (5–6): 507–517. DOI: 10.1016/S0145-305X(98)00030-5
- Seto F. 1981. Early development of the avian immune system. *Poultry Sci.* **60** (9): 1981–1995. DOI: 10.3382/ps.0601981
- Sharma J.M. 1997. The structure and function of the avian immune system. *Acta Vet. Hung.* **45** (3): 229–238. PMID: 9276985
- Síma P. 1997. The development of the defence system during evolution, pp. 6–20. In: Heidt P.J., Rusch V. & der Waaij D. (eds), Old Herborn University Seminar Monograph, 10. New

- Antimicrobial Strategies, Herborn Litterae, Herborn-Dill, 137 pp. ISBN: 3-923022-20-4
- Sima P. & Vetricka V. 1993. Evolution of immune reactions. *Crit. Rev. Immunol.* **13** (2): 83–114. PMID: 8352910
- Slipka J. 1986. Evolutionary morphology of the branchial region as the reflection of environmental changes, pp. 203–211. In: Novák V.J.A., Vančata V. & Vančatová M.A. (eds), *Behaviour, Adaptation and Evolution*, Czechoslovak Academy of Sciences, Praha.
- Slipka J. & Slipka J.Jr. 1996. The palatine tonsil as an evolutionary novelty. *Acta Otolaryngol. Suppl.* **523**: 8–11. PMID: 9082817
- Smith K.G. & Hunt J.L. 2004. On the use of spleen mass as a measure of avian immune system strength. *Oecologia* **138** (1): 28–31. DOI: 10.1007/s00442-003-1409-y
- Spalding H. & Heath T. 1987. Pathways of lymph flow through superficial inguinal lymph nodes in the pig. *Anat. Rec.* **217** (2): 188–195. DOI: 10.1002/ar.1092170211
- Sypek J. & Borysenko M. 1988. Reptiles. In: Rowley A.F. & Ratcliffe N.A. (eds), *Vertebrate Blood Cells*, Press Syndicate in the University of Cambridge, 432 pp. ISBN: 0-521-26032-9
- Štěrba G. 1953. Die Physiologie und Histogenese der Schilddrüse und des Thymus beim Bachneunauge (*Lampetra planeri* Bloch = *Petromyzon planeri* Bloch) als Grundlagen phylogenetischer Studien über die Evolution der innersekretorischen Kiemendarmderivate nebenst eingehenden Mitteilungen über die Bionomie der Bachneunaugen und morphologisch-physiologischen Untersuchungen über die Kiemendarm. *Wissenschaftliche Zeitschrift. Mathematisch-Naturwissenschaftliche Reihe* **3** (2): 239–298.
- Terszowski G., Muller S.M., Bleul C.C., Blum C., Schirmback R., Reimann J., Pasquier L. D., Amagai T., Boehm T. & Rodewald H.R. 2006. Evidence for a functional second thymus in mice. *Science* **312** (5771): 284–287. DOI: 10.1126/science.1123497
- Tischendorf F. 1985. On the evolution of the spleen. *Experientia* **41** (2): 145–152. PMID: 3972062
- Tochinai S. 1976. Lymphoid changes in *Xenopus laevis* following thymectomy at the initial stage of its histogenesis. *J. Fac. Sci. Hokkaido Univ. Ser. Zool.* **20** (2): 175–184.
- Valančiūtė A., Mozuraitė R., Balnytė I., Didžiapetrienė J., Matusevičius P. & Stakišaitis D. 2015. Sodium valproate effect on the structure of rat glandule thymus: Gender-related differences. *Exp. Toxicol. Pathol.* **67** (7-8): 399–406. DOI: 10.1016/j.etp.2015.04.005
- Valasek P., Macharia R., Neuhuber W.L., Wilting J., Becker D.L. & Patel K. 2007. Lymph heart in chick—somitic origin, development and embryonic oedema. *Development*. **134**(24): 4427–4436. DOI: 10.1242/dev.004697
- Van de Pavert S.A. & Mebius R.E. 2010. New insights into the development of the lymphoid tissue. *Nat. Rev. Immunol.* **10** (9): 664–674. DOI: 10.1038/nri2832
- Varas A., Sacedón R., Hernandez-López C., Jiménez E., García-Ceca J., Arias-Díaz J., Zapata A.G. & Vicente A. 2003. Age-dependent changes in thymic macrophages and dendritic cells. *Microsc. Res. Techn.* **62** (6): 501–507. DOI: 10.1002/jemt.10411
- Varga I., Galfiova P., Jablonska-Mestanova V., Polak S. & Adamkov M. 2011a. Some aspects of early development of the thymus: Embryological basis for ectopic thymus and thymopharyngeal duct cyst. *Rev. Arg. Anat. Clin.* **3** (1): 22–31.
- Varga I., Mikusova R., Pospisilova V., Galfiova P., Adamkov M., Polak S. & Galbavy S. 2009. Morphologic heterogeneity of human thymic nonlymphocytic cells. *Neuro Endocrinol. Lett.* **30** (3): 275–283. PMID: 19855349
- Varga I., Nescakova E., Toth F., Uhrinova A. & Adamkov M. 2011b. Nutrition and immune system: the size of the thymus as an indicator of the newborn's nutrition status. *Anthropol. Anz.* **68**(3): 265–274. PMID: 21905416
- Varga I., Pospisilova V., Gmitterova K., Galfiova P., Polak S. & Galbavy S. 2008. The phylogenesis and ontogenesis of the human pharyngeal region focused on the thymus, parathyroid, and thyroid glands. *Neuroendocrinol. Lett.* **29** (6): 837–845. PMID: 19112385
- Varga I., Pospisilova V., Jablonska V., Sisovsky V., Galfiova P., Polak S. & Adamkov M. 2010. Thymic Hassall's bodies of children with congenital heart defects. *Bratisl. Lek. Listy* **111** (10): 552–557. PMID: 21125801
- Varga I., Pospisilova V., Jablonska-Mestanova V., Galfiova P. & Polak S. 2011c. The thymus: picture review of human thymus prenatal development. *Bratisl. Lek. Listy* **112** (7): 368–376. PMID: 21744730
- Vasse J. 1983. Transplantation of turtle embryonic thymus into quail embryo: colonization by quail cells. *J. Embryol. Exp. Morph.* **77**: 309–322. PMID: 6606698
- Vigliano F.A., Losada A.P., Castello M., Bermúdez R. & Quiroga M.I. 2011. Morphological and immunohistochemical characterisation of the thymus in juvenile turbot (*Psetta maxima* L.). *Cell Tissue Res.* **346** (3): 407–416. DOI: 10.1007/s00441-011-1282-7
- Wakimoto T., Tomisaka R., Nishikawa Y., Sato H., Yoshino T. & Takahashi K. 2008. Identification and characterization of human thymic cortical dendritic macrophages that may act as professional scavengers of apoptotic thymocytes. *Immunobiol.* **213** (9-10): 837–847. DOI: 10.1016/j.imbio.2008.07.032
- Wetherall J. D. & Turner K. J., 1972. Immune response of the lizard, *Tiliqua rugose*. *Aust. J. Exp. Biol. Med. Sci.* **50**: 79–95. DOI: 10.1038/icb.1972.7
- White R.G. 1976. Organization of the lymphoid tissue of *Galus domesticus*, pp. 15–29. In: Payne L.N. (ed.), *Differential Diagnosis of Avian Lymphoid Leucosis and Marek's Disease*, Commission of the European Communities. Coordination of Agricultural research. Proceedings of a Seminar in the EEC Programme for Coordination of Research on Avian Leukosis held at the Royal Veterinary and Agricultural University of Copenhagen, Copenhagen, Denmark, Directorate-General, Scientific and Technical Information and Information Management, Luxembourg 99 pp.
- Wolke R.E. 1992. Piscine macrophage aggregates: A review. *Annu. Rev. Fish Dis.* **2**: 91–108. DOI: 10.1016/0959-8030(92)90058-6
- Wong E.S., Papenfuss A.T., Heger A., Hsu A.L., Ponting C.P., Miller R.D., Fenelon J.C., Renfree M.B., Gibbs R.A. & Belov K. 2011. Transcriptomic analysis supports similar functional roles for the two thymuses of the tammar wallaby. *BMC Genomics* **12**: 420. DOI: 10.1186/1471-2164-12-420
- Yasuda M., Jenne C.N., Kennedy L.J. & Reynolds J.D. 2006. The sheep and cattle Peyer's patch as a site of B-cell development. *Vet. Res.* **37** (3): 401–415. DOI: 10.1051/vetres:2006008
- Zaitseva M., Kawamura T., Loomis R., Goldstein H., Blauvelt A. & Golding H. 2002. Stromal-derived factor 1 expression in the human thymus. *J. Immunol.* **168** (6): 2609–2617. DOI: 10.4049/jimmunol.168.6.2609
- Zapata A. & Amemiya C.T. 2000. Phylogeny of lower vertebrates and their immunological structures. *Curr. Top. Microbiol. Immunol.* **248**: 67–107. DOI: 10.1007/978-3-642-59674-2_5
- Zapata A., Diez B., Cejalvo T., Gutiérrez-de Frías C. & Cortés A. 2006. Ontogeny of the immune system of fish. *Fish Shellfish Immunol.* **20** (2): 126–136. DOI: 10.1016/j.fsi.2004.09.005
- Zapata A.G., Torroba M., Vicente A., Varas A., Sacedón R. & Jiménez E. 1995. The relevance of cell microenvironments for the appearance of lympho-haemopoietic tissues in primitive vertebrates. *Histol. Histopathol.* **10** (3): 761–778. PMID: 7579826
- Zapata A., Villena A. & Cooper E.L. 1981a. Ultrastructure of the jugular body of *Rana pipiens*. *Cell Tissue Res.* **221** (1): 193–202. DOI: 10.1007/BF00216581
- Zapata A., Villena A., Razquin B. & Cooper E.L. 1981b. The jugular body in anuran amphibians: role in immunity. *Dev. Comp. Immunol.* **5** (Suppl. 1): 129–135.
- Zidan M. & Pabst R. 2010. Histology of hemal nodes of the water buffalo (*Bos bubalus*). *Cell Tissue Res.* **340** (3): 491–496. DOI: 10.1007/s00441-010-0962-z

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